RSRT-DETR: Hierarchical Polar Attention and Multiscale Hypergraph Networks for Dense Remote Sensing Small-Object Detection

Hongyang Zhao, *Member, IEEE*, Kai Chen, Yao Zhang, Xingdong Li, Honggang Li, and Jing Jin

***Abstract*—Recent advances in optical remote sensing have expanded the use of uncrewed aerial vehicles (UAVs), yet accurately localizing dense, small objects in low-resolution UAV imagery remains challenging. We propose remote sensing real-time detection transformer (RSRT-DETR), a lightweight DETR derivative tailored for UAV-borne remote sensing that jointly enhances global context modeling and fine-grained detail preservation. RSRT-DETR couples a polar dynamic spatial fine-tuning (PDSF) attention module with a lightweight backbone to amplify fine-grained geometry while suppressing background noise, and embeds a multiscale hypergraph feature network (MS-HFNet) that reasons over high-order spatial relations to disambiguate cluttered targets. These features are hierarchically fused by the hypersemantic fusion network (HSFN), whose progressive edge-enhancement, resolution-adaptation, and multi-granularity aggregation schemes jointly preserve cross-scale context without sacrificing speed. Evaluated on five challenging benchmarks—DOTAv1.0, VisDrone2019, NWPU VHR-10, DIOR, and HIT-UAV—RSRT-DETR achieves superior performance, with its AP_{50} surpassing state-of-the-art methods and its AP_S reaching the highest value. Meanwhile, the number of parameters and FLOPs is lower than those of similar DETR variants. The above results prove that RSRT-DETR achieves the optimal coordination of detection accuracy, model scale, and real-time performance in complex backgrounds and multiscale dense small-target scenes, providing a new paradigm for building efficient and reliable UAV remote sensing target detection systems. Our code is available on: https://github.com/zhy1109/RSRT-DETR**

***Index Terms*—Remote sensing real-time detection transformer (RSRT-DETR), object detection, remote sensing, small-target detection.**

I. Introduction

IN RECENT years, with the rapid development of optical remote sensing technology, the research on remote sensing images has achieved significant growth [1], [2], [3], [4], [5]. As an important tool for target detection in the remote sensing field, UAVs are widely used in many fields, such as transportation [6], rescue [7], safety [8], and military, due to their high maneuverability, broad vision, low cost, and convenient deployment. Compared with other remote sensing object detection tools, the main task of drone detection is to identify and locate objects in the images it captures and videos. However, in the images captured by the drone, the lower resolution and poor image quality pose many difficulties to target recognition [9]. The captured images show blurred features, lower contrast, and a lack of information, while the targets of the model's interest are often smaller [10], [11]. These factors greatly increase the difficulty of detection [12]. In addition, during the imaging process, the imaging link faces many uncontrollable factors such as the movement of the platform, interference from the atmospheric environment, and complex and changeable imaging scenes. These will cause mutual aliasing between objects and backgrounds, making it more difficult to identify small-sized objects [13]. At the same time, since the captured remote sensing images often carry a large amount of redundant information, it not only increases the processing burden but also leads to a decrease in recognition accuracy and efficiency.

With the rapid advancement of deep learning technology in the past few years, researchers have begun to try to combine it with drone detection to solve the above problems. These methods are based on a convolutional neural network (CNN). Through an end-to-end learning framework and multiscale feature fusion mechanism, they achieve accurate image recognition and effectively improve the performance of drone target detection. In particular, deep learning models, such as Faster R-CNN [14], SSD [15], and YOLO [16] have achieved excellent performance on common object detection datasets, and are therefore widely used in UAV target detection. However, such models are based on the inherent characteristics of convolutional operations, resulting in confinement of receptive fields, making it difficult to model spatial correlations between long-distance targets. At the same time, when faced with zero complex geographical environment and lighting conditions, diversity of target scales, and dense target occlusion, the detection accuracy of this type of model is still insufficient. Recently, the remote sensing real-time detection transformer (RSRT-DETR) detection model based on the Transformer architecture has begun to be applied to drone detection tasks, providing new ideas for breaking through the above bottlenecks [17]. Where RT-DETR only applies the self-attention mechanism to high-level semantic rich feature maps and integrates it with convolutional operations to perform multiscale

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feature fusion. This significantly improves the global modeling capability of the model and effectively reduces computational redundancy. The design improves detection speed while maintaining high accuracy, proposing a completely new solution for the UAV detection model. However, there are still several key bottlenecks in this type of detection model. First, in drone detection tasks, the image area of small targets accounts for a small amount, and the spatial resolution of high-level feature maps is insufficient, which will lead to a serious loss of geometric details of small targets. Second, existing methods mostly use fixed-scale attention windows [18], making it difficult to dynamically adapt to the drastic changes in the target scale captured by drones at different flight altitudes. In addition, although this type of model has achieved high detection accuracy and computing overhead, it has not yet achieved a good balance in these two aspects, which poses a challenge to the actual deployment needs of the model on drones.

To solve the above problem, we propose a new DETR variant, RSRT-DETR, to improve the recognition and positioning performance of small targets in complex contexts in UAV detection tasks. As shown in Fig. 1, the unique design of RSRT-DETR can effectively reduce the missed detection rate and false detection rate of the model, while also effectively balancing the recognition accuracy and recognition efficiency. To verify the performance of RSRT-DETR, we conducted experiments on three challenging small-target datasets. Our model achieves the best performance compared to the current mainstream approaches and the most advanced models. It provides a new paradigm for building efficient and reliable drone detection systems. The main contributions of this article are as follows.

1) We propose a brand new DETR variant: RSRT-DETR, which is used for remote sensing image detection. RSRT-DETR is particularly sensitive to small targets and multiscale targets, and at the same time, it can achieve higher detection accuracy under low computational volume, making it more suitable for UAV detection tasks.
2) Based on the self-attention mechanism, we propose a polar dynamic spatial fine-tuning attention (PDSF Attention) mechanism. By decomposing traditional self-attention into dual paths, combining dynamic Tanh normalization and Mona adapter, the defects of traditional Transformer's redundancy in global attention calculation and insufficient local detail response in complex remote sensing scenarios are effectively solved.
3) In order to solve the problem of feature ambiguity and context separation in multiscale object detection, we designed a multiscale hypergraph feature network (MS-HFNet). By modeling the high-order spatial correlation between feature nodes through hypergraphs and combining multibranch parallel processing, the network's perception ability of multiscale semantics and edge information is effectively improved.
4) We designed a progressive edge semantic enhanced fusion architecture, hypersemantic fusion network (HSFN), and proposed three modules to solve different challenges: multigranularity semantic aggregation layer (MGSAL) enhances multiscale feature fusion between different layers, where adaptive hierarchical context edge fusion (AH-CEF) realizes feature interactions between adjacent levels; attention-guided resolution adaptation (ARGA) optimizes resolution variation problems existing between different levels; and progressive edge strengthening module (PESM) improves interscale correlations between network contexts. The design of these three core modules enables HSFN to effectively alleviate interference from complex backgrounds at different scales in complex environments, effectively improving the robustness of the model.

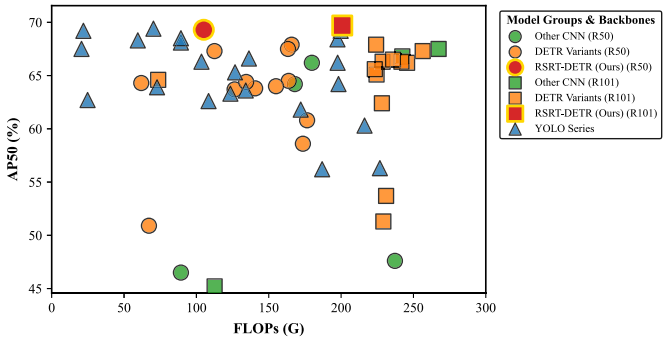


Fig. 1. Comparative experiments with other mainstream methods and SOTA models on the DOTA dataset.

II. Related Works

A. Object Detection in Remote Sensing Based on UAV

As an important part of remote sensing detection technology, drone remote sensing target detection technology has shown important value in disaster monitoring, traffic management, and other fields in recent years. With the rapid advancement of computer vision and deep learning, more and more researchers are beginning to apply deep learning models to UAVs to achieve remote sensing monitoring tasks. Object detection models based on deep learning can generally be divided into two categories: object detection models based on CNN architecture [19], [20], [21], [22], [23] and object detection models based on Transformer architecture [20], [24], [25], [26], [27].

The object detection model based on the CNN architecture extracts the target features through local convolution operations and hierarchical feature pyramids, and uses the spatial translation invariance of the convolution kernel to capture local-to-global feature information step by step. Thanks to its ability to quickly extract local features, its detection efficiency is high and has been widely used in UAV detection. In response to the challenge of confusion of complex backgrounds in images, FFCA-YOLO [28] optimizes the fusion of multiscale semantics through channel reweighting strategies, and utilizes the correlation between global pooling resumes across channels and space to effectively suppress interference from complex backgrounds. TLSTMF-YOLO [29] combines the bidirectional feature pyramid network to achieve cross-scale features, and at the same time, the introduction of

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transfer learning strategies has solved the problems of overfitting and convergence efficiency in small sample scenarios. LSKF-YOLO [30] starts from another perspective and builds a hierarchical context-aware network by combining the large-space selection core mechanism and attention feature fusion module, effectively improving the model's detection performance of targets in complex contexts.

Although the object detection model based on the CNN architecture has achieved a high detection speed, due to the limitation of the receptive field brought about by the convolution operation, it is difficult to capture global features, resulting in a significant reduction in its recognition performance in complex backgrounds. In contrast, the object detection model based on the Transformer architecture models long-range dependencies through the global self-attention mechanism, directly establishes interrelationships between different regions, and achieves accurate positioning of dense targets in complex scenarios. In addition, its powerful global modeling ability and excellent feature capture ability can enable it to effectively extract the features of dense targets in complex scenarios without requiring anchor frames and postprocessing. Therefore, the object detection model based on the Transformer architecture is more suitable for use in complex environmental backgrounds and intensive small-objective tasks.

B. Object Detection in Small-Object

In the remote sensing images captured by UAV, the number of small objects is large, and the proportion of the entire image is very small, which leads to the fact that their characteristics may occupy very few pixels on the output feature map, which is a key factor in the missed and misdetection of the detection model. Therefore, cross-scale fusion of multiscale information extracted from the backbone network is a necessary condition to improve the accuracy of small-object detection. Lin et al. [31] first proposed a feature pyramid and used it to fuse shallow and deep features. On this basis, EfficientDet [32] proposed BiFPN, which introduced bidirectional connections between different levels of the pyramid, enabling information to flow from bottom to top and top to bottom to effectively improve information interaction. Swin Transformer [33] starts from the other hand and uses the window slip mechanism in the self-attention mechanism to improve the model's perception of local details and global features, effectively improving the model's sensitivity to small-target features. DINO-DETR [34] proposes a two-way interactive attention and multiscale enhancement module and is used in the neck network, effectively improving the model's detection performance on small targets. Overlock [35] designs a branch architecture based on traditional pyramid networks and establishes three collaborative subnets to process image features in parallel, effectively improving the model's refined fusion ability of small-target features. These results show that a correct combination of multiscale information can effectively improve the detection performance of the model for small-object detection. In addition, some scholars have tried to break through the traditional image fusion paradigm and have achieved good results. Hyper-YOLO [36] breaks through the learning strategy that models can only fit through image features by mapping traditional visual features into semantic space and building hypergraphs for information propagation, and allowing complex high-order interactions across layers and locations.

C. DETR and Its Variants

As an end-to-end object detection model based on Transformer, DETR, compared with other object detection models based on Transformer architecture, abandons the nonmaximum suppression in traditional anchor box design and postprocessing steps, and adopts a design that matches ensemble prediction and binary graphs. It directly models input features through global attention, showing strong performance in object detection in complex scenarios. Therefore, the researchers improved it and developed many variants: deformable DETR [37] realizes dynamic selection of feature positions by introducing a deformable attention mechanism while adopting a multilevel feature fusion strategy to reduce the computational overhead of DETR and improve convergence speed. On this basis, SED-DETR [38] dynamically adjusts the feature information of different scales based on the principles of deformable shape and dynamic scale, effectively enhancing the model's scale perception modeling ability. In terms of spatial prior modeling, DAB-DETR [39] innovatively uses dynamic anchor box parameters as position code during query, allowing the cross attention module to focus on the potential areas of the target object, significantly enhancing the accuracy of detection and positioning. Anchor-DETR [40] further deepens the design idea of position priors, by explicitly defining anchor points at different scales as initial queries, and combining adaptive sparse sampling strategies, it improves the sensitivity of small-object detection while reducing calculation redundancy. In response to the computational efficiency problem, DN-DETR [41] proposed a denoising training paradigm. By designing auxiliary supervision tasks containing coordinate perturbations, the convergence speed and detection robustness of the model are effectively improved while bypassing binary graph matching. Focus-DETR [42] proposes a dual-path feature focusing mechanism from the perspective of attention feature focusing, and achieves dynamic screening of target-related features through significance weight prediction, and designs a spatially sensitive mask to suppress background noise interference. PSWP-DETR [43] uses adaptive learning, horizontal, and vertical deformation to perceive the complex geometric features of small objects in complex backgrounds, and at the same time extracts edge features through the principle of scale difference, effectively improving the convergence speed of the model. In order to solve the detection robustness problem, SC-RTDETR [44] significantly enhances the recall of the model by introducing dynamic threshold adjustments in the feature extraction phase and using a grouped cascade attention mechanism in the encoder. ARS-DETR [45] uses SkewIoU to dynamically adjust the label smoothing strategy, rotates the sampling points according to the predicted angle to solve the misalignment problem between the characteristics and the sampling points, effectively improving the detection performance.

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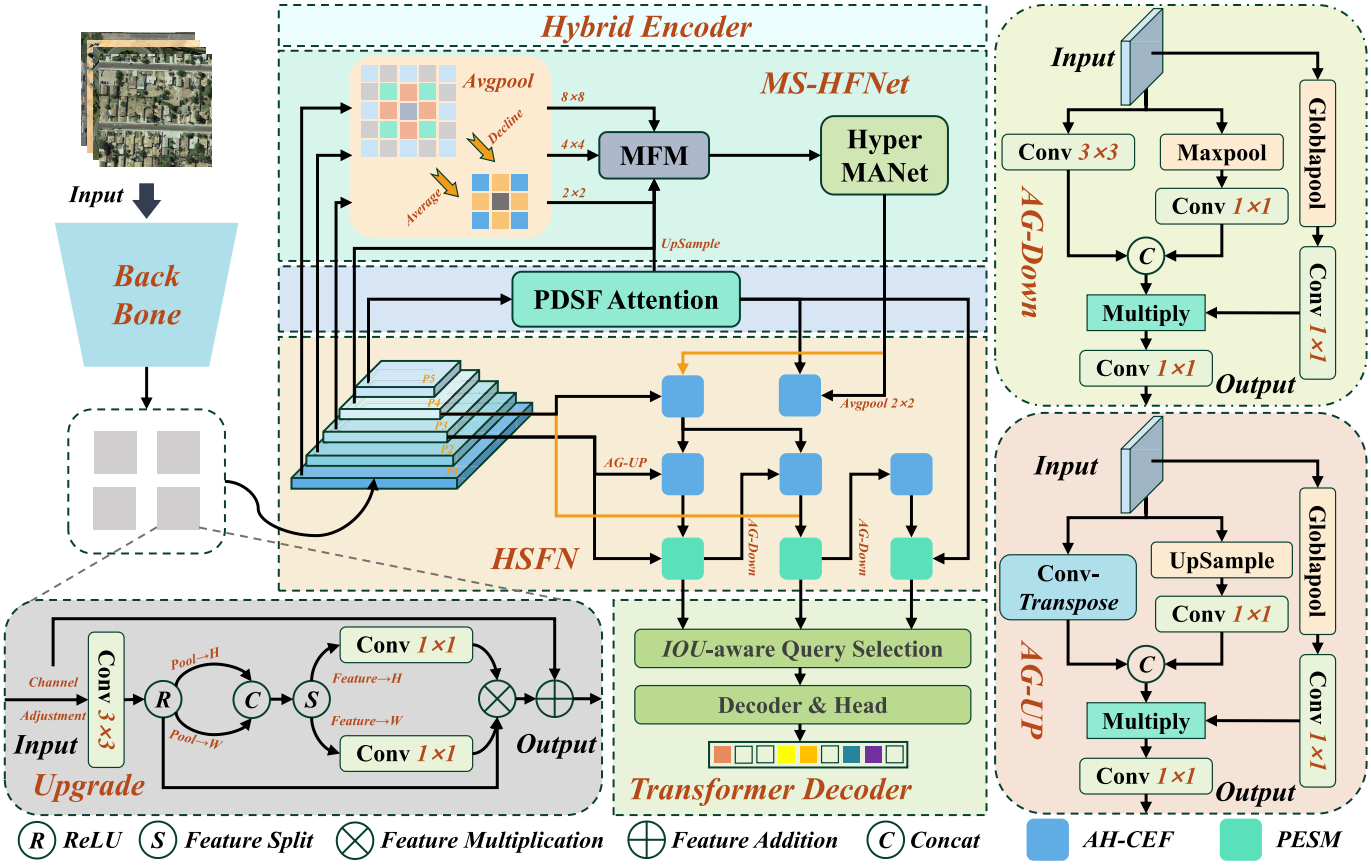


Fig. 2. Overall architecture of RSRT-DETR, which contains three key components: backbone network, encoder, and decoder. After passing through the backbone network, the extracted multiscale features are fed into the cross-feature pyramid and unpacked into three paths, one of which is directly connected to MS-HFNet. Another path passes through the PDSF Attention before being delivered to the MS-HFNet. Another path is sent to HSFN with MS-HFNet and PDSF Attention. Next, the features output by the HSFN are entered into the IoU-aware for query selection, and the decoder is used for the initial object to query initialization. Finally, the decoder in the auxiliary prediction header iterates to update the bounding box generated by the model and the confidence scores for each category.

III. Methodology

A. Overall Framework of RSRT-DETR

The network architecture of RSRT-DETR is shown in Fig. 2, which mainly consists of three parts: feature extraction network, hybrid encoder, and transformer decoder. First, the input image is downsampled through the feature extraction network to generate feature information of different scales. Inspired by RT-DETR [46], we designed a new multiscale feature encoder. Since the deep features (P5) extracted by the feature extraction network contain rich semantic information, we sent them to PDSF Attention after using channel compression for feature enhancement, and conveyed them to MS-HFNet as the core part of subsequent multiscale fusion. MS-HFNet can not only realize feature interactions at multiple scales, but also deeply mine discriminant details between small objects and complex backgrounds in remote sensing images, effectively enhancing the network's ability to characterize small details. After integrating effective features, RSRT-DETR uses HSFN to effectively integrate features, achieving cross-scale fine interaction between multiscale features. In the feature decoder stage, we encode the fused multilevel features into a dynamic target query, and generate the object detection box through the Transformer driver, thereby achieving the one-to-one correspondence between the predicted bounding box and the ground real tag.

B. PDSF Attention

For remote sensing image detection models, how to achieve precise positioning of dense small targets under complex background interference is their core challenge. However, the traditional Transformer encoder uses a global self-attention mechanism and serialized position coding. Although it can model long-distance dependencies, it faces high computational overhead and weak local details response capabilities. In order to solve these problems, we propose PDSF Attention based on the self-attention mechanism, and achieve coordinated optimization of global semantic modeling and local detail enhancement through multistage hybrid operations. As shown in Fig. 3, for deep feature $R^{B\times C\times H\times W}$, PDSF Attention first uses a perceived attention mechanism for feature enhancement. Specifically, subchannel polarization attention decomposes traditional self-attention into similar paths and opposite paths, and uses dynamically learnable nonlinear kernel parameters to polarize spatially map the Q and K elements in self-attention. The similar path strengthens the consistency characteristics of

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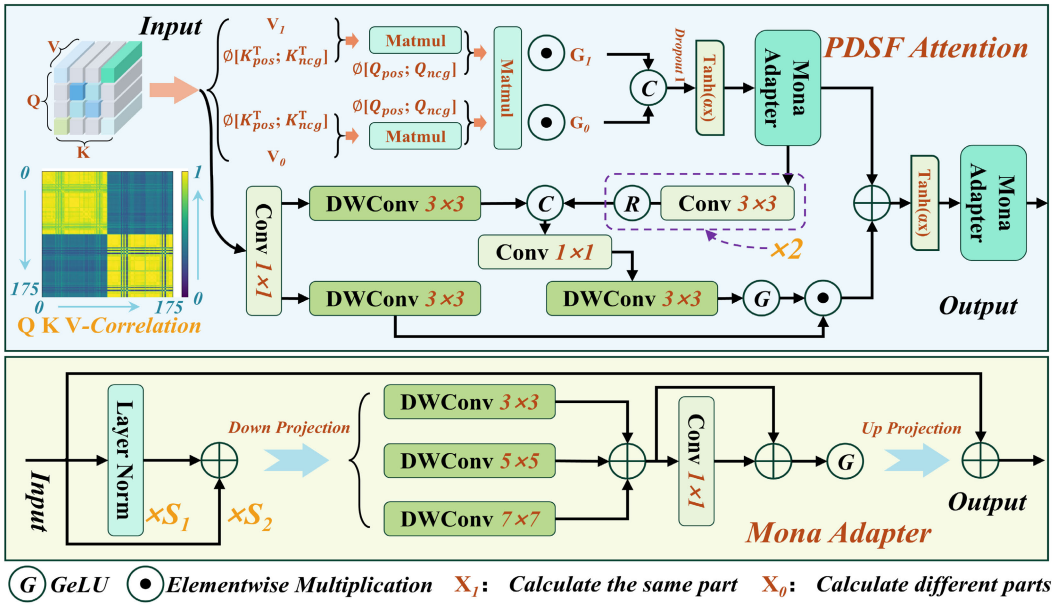


Fig. 3. Complete architecture of the PDSF Attention.

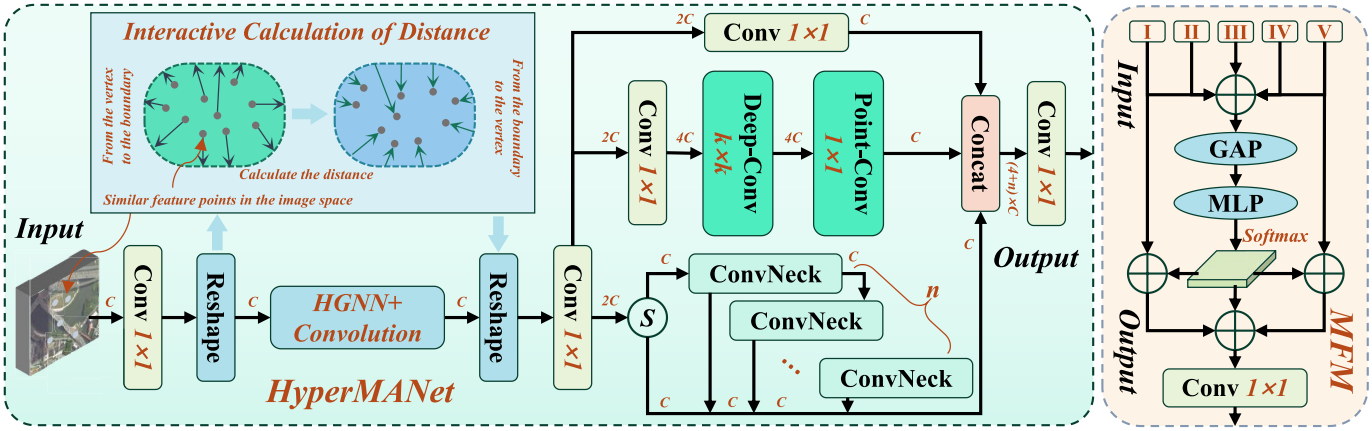


Fig. 4. Overall framework of HyperMANet and the specific flowchart of MFM module.

the target area, and the opposite path suppresses interference from background noise. Specifically, it is expressed as follows:

$$A_{sim} = Softmax\left(\frac{(Q_{pos} \cdot K_{pos}^{\top})}{\sqrt{D}}\right) \cdot V_1 \quad (1)$$

$$A_{opp} = Softmax\left(\frac{(Q_{ncg} \cdot K_{ncg}^{\top})}{\sqrt{D}}\right) \cdot V_2 \quad (2)$$

where A_{sim} represents a similar path, A_{opp} represents an opposite path, $D = C/H$ represents the dimension of each attention head, V_1 and V_2 represent the positive and negative semantic features, respectively, Q_{pos} and Q_{ncg} represent the similar and opposite features polarized by Q, and $K_{pos}^{\top}$ and $K_{ncg}^{\top}$ represent the similar and opposite features polarized by K and transpose matrix of similar and opposite features, they satisfy both as

$$X_{pos} = ReLU\left((X_{raw})\right)^{Sigmoid(\alpha)+1} \quad (3)$$

$$X_{ncg} = ReLU\left(((-X)_{raw})\right)^{Sigmoid(\alpha)+1} \quad (4)$$

where $\alpha \in R^{1\times H\times 1\times D}$ is a learnable dynamic kernel parameter, and X_{raw} represents the original feature. The features generated by similar paths and opposite paths are finally fused through depth separation convolution and then weighted to the dynamic normalization layer. The dynamic Tanh normalization layer dynamically adjusts the feature distribution range through channel adaptive scaling factors and bias terms to enhance the robustness of the model for complex lighting and background. Specifically, it can be expressed as follows:

$$Norm(x) = \tanh(\alpha \cdot x) \odot W + b + \gamma \cdot LayerNorm(x) \quad (5)$$

where α represents the learnable scaling factor that controls Tanh activation intensity, while $W, b \in R^{C\times 1\times 1}$ is the channel adaptive scaling and bias parameters, and c represents the dynamic weight parameters. Subsequently, the spatially enhanced feedforward network is processed through two parallel branches: the main branch and the multiscale pooling–upsampling branch. Multiscale pooling–upsampled branches extract cross-resolution context information and

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Algorithm 1 PDSF Attention

Require: $X \in \mathbb{R}^{B\times C\times H\times W}$ ▷2-D feature map $\theta = \{\mathbf{W}_A, \mathbf{W}_F, \mathbf{W}_N, \mathbf{W}_M, \ldots\}$ ▷learnable weights $\text{drop}_1, \text{drop}_2$ ▷dropout ratios

Ensure: $Y \in \mathbb{R}^{B\times C\times H\times W}$ ▷ output feature map

Procedure FORWARD(X)

$X_a \leftarrow$ PolaAttention(FlattenHW(X)) ▷global polar self-attention

$X_a \leftarrow$ Reshape(X_a, B, C, H, W)

$X \leftarrow X + \text{Dropout}(X_a, \text{rate} = \text{drop}_1)$ ▷first residual

$X \leftarrow \text{DynamicTanhNorm}_1(X)$ ▷post-norm (DyT)

$X \leftarrow \text{Mona}_1(X)$ ▷multikernel locality mixing

$X_{FF} \leftarrow$ SefnFeedForward($X, \text{context} = X$) ▷selective expansion FFN

$X \leftarrow X + \text{Dropout}(X_{FF}, \text{rate} = \text{drop}_2)$ ▷second residual

$X \leftarrow \text{DynamicTanhNorm}_2(X)$ ▷post-norm (DyT)

$Y \leftarrow \text{Mona}_2(X)$ ▷output modulation

return Y

End Procedure

gating and fuse with the deep convolutional features of the main branch, using a dynamic gating mechanism to achieve semantic-detail balance. In this process, in order to reduce the computational overhead, we use the Mona adapter to further fine-tune the feature space through lightweight convolution and parameterized residual connection to achieve lightweight feature calibration. In order to clearly indicate the working mechanism of the PDSF Attention module, the calculation process is outlined in Algorithm 1. The entire process always maintains a 2-D feature structure to avoid information losses caused by explicit position coding and serialization operations. Compared with traditional Transformer encoders, the features output by PDSF Attention contain more feature details, ensuring that the model still has a high sensitivity to small targets in complex scenarios.

C. Multiscale Hypergraph Feature Network

In order to effectively improve the perception ability of the DETR model on multiscale semantics and edge information, we propose MS-HFNet. By constructing a hierarchical dynamic fusion and edge-aware optimization mechanism, MS-HFNet can effectively fuse features of different scales, significantly improving the model's feature discrimination ability of multiscale targets in complex scenarios. As shown in Fig. 4, Since the deep information in the backbone network is characterized by PDSF Attention, in order to make full use of the remaining shallow information in the backbone network, MS-HFNet applies average pooling of different core sizes to extract global semantic information of different granularity, and inputs the extracted shallow features with the PDSF Attention enhanced P5 features and upsampled high-resolution features to the multifeature mixing module (MFM) for feature fusion. MFM adopts a channel attention mechanism, dynamically generates feature weights through global average pooling and multilayer perceptrons, achieving cross-level semantic adaptive fusion. For the input feature set $[F_k]_{k=1}^N \in R^{B\times C_k\times H\times W}$, MFM first performs dimension unification

$$F_k' = \begin{cases} \text{Conv}_{1\times 1}(F_k), & \text{if } C_k \neq C \\ F_k, & \text{otherwise.} \end{cases} \tag{6}$$

After the channel alignment, the feature dimensions are consistent with $[F_k']_{k=1}^N \in R^{B\times C_k\times H\times W}$, and are then spliced along the channel dimension and reshaped into 3-D tensors. After this process, the MFM sums the features of each spatial position, compresses it into a global description vector through global average pooling, and inputs multilayer perceptrons to generate channel attention weights for each input feature. Finally, multiply each input feature with the corresponding attention weights and sum it by channel to output the fused feature

$$F_{fused} = \sum_{k=1}^{N}\left(Softmax\left(MLP\left(GAP\left(\sum_{i=1}^{N} F_i\right)\right)\right)\right)\bigodot F_k \tag{7}$$

where MLP adopts a dimensionality reduction design, that is, the first layer compresses the number of channels to $C/8$, the second layer restores to $C \times N$, and generates a weight matrix corresponding to the number of input features N. This mechanism can effectively suppress background interference and enhance the response of key areas. In order to solve the problem of feature ambiguity between dense targets, MS-HFNet innovatively proposed HyperMANet. By modeling high-order spatial correlations between features at different scales, it can effectively improve the geometric accuracy of single targets while also improving the robustness of correlation modeling between targets. For the input feature $X \in R^{B\times C\times H\times W}$, HyperMANet first models the global spatial association through a hypergraph: reshape the features into a node sequence $X_{flat} \in R^{B(HW)C}$ and define $H \times W$ nodes, and then calculate the Euclidean distance between all node pairs to form a distance matrix D

$$D_{i,j} = \sqrt{\sum\nolimits_{c=1}^{C}\left(X_{flat}^{(i,c)} - X_{flat}^{(j,c)}\right)^2}. \tag{8}$$

In order to control hyperedge connections, we set the threshold τ. If $D_{i,j} < \tau$, we define nodes i and j belong to the same hyperedge and generate a binary adjacency matrix $H \in [0,1]^{B\times(H\times W)\times(H\times W)}$. Then, we perform bidirectional message delivery, which includes two stages: node → hyperedge and hyperedge → node. In the node → hyperedge stage, each hyperedge aggregates the characteristics of its connected nodes, and uses mean normalization to avoid gradient instability

$$E = LayerNorm\frac{H^T \cdot (X_{flat}W_{proj})}{\sum H^T + \epsilon},\ E \in R^{B\times N_{hyper}\times C} \tag{9}$$

where $W_{proj} \in R^{C\times C}$ is the learning linear transformation matrix, N_{hyper} is the super edge number, and ϵ is the minimum value to prevent zero division. In the hyperedge → node stage, the hyperedge feature is backpropagated to the node

$$X_{update} = \frac{H \cdot E}{\sum H + \epsilon},\ (X)_{update} \in R^{B\times(H\times W)\times C}. \tag{10}$$

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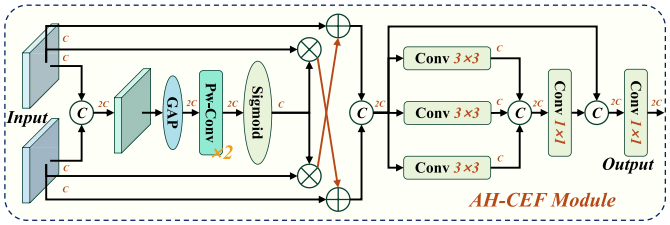


Fig. 5. Structure of AH-CEF module.

The final two stages generate the hypergraph feature X_{hyper} through the residual connection. Then, the hypergraph features are sent to the multibranch attention network again and split into four channels for parallel processing

$$Y_{path1} = Bottleneck(X_{out}) \tag{11}$$

$$Y_{path2} = DWConv(X_{out}) \tag{12}$$

$$Y_{path3} = (Conv)_{3\times3}^{d=2}(X_{out}) \tag{13}$$

$$Y_{path4} = X_{out}. \tag{14}$$

Among them, branch 1 adopts a bottleneck structure to extract local details, branch 2 uses depth separation convolution to lighten the context, branch 3 uses 3×3 convolution with expansion rate $d = 2$ to cover larger areas, capture the intermediate-range context, and branch 4 retains the original features to prevent information loss. Finally, HyperMANet splices the four branch outputs and obtains the fused feature Y_{fused} through 1×1 convolution

$$Y_{fused} = Conv_{1\times1}\left(Concat\left(Y_{path1}, Y_{path2}, Y_{path3}, Y_{path4}\right)\right). \tag{15}$$

Compared with traditional FPN structures, MS-HFNet innovatively integrates the advanced relational reasoning of hypergraphs with multibranch local–global feature extraction capabilities. In remote sensing image detection tasks that detect targets and complex backgrounds, it can simultaneously improve the robustness of correlation modeling between targets and single-objective geometric accuracy, and maintain real-time performance through channel compression and computing sharing.

D. Hypersemantic Fusion Network

In order to make full use of the feature information generated by the backbone network while overcoming the context splitting and edge blur problems in small-object detection, we abandoned the traditional one-way feature fusion method and proposed a new HSFN, and achieved refined interaction of cross-scale features through multistage semantic enhancement and edge optimization mechanisms. The core process is described below.

1) Multigranularity Semantic Aggregation Layer: In the feature refining stage, the input features are initially refined in MGSAL. MGSAL first performs average pooling and dimensionality reduction on the deep feature P5 with the size 2×2, and performs bidirectional attention compensation with the shallow feature through the AH-CEF module. As shown in Fig. 5, for the input adjacent hierarchical features $F_{low} \in R^{C\times H\times W}$ (low-layer high resolution) and $F_{high} \in R^{C\times H/2\times W/2}$ (high-layer low resolution), AH-CEF first uses 1×1 convolution to align the channel dimensions and obtains the dimension reconstruction feature $F_{cat} = Concat(F_{align}, F_{high}) \in R^{2C\times H/2\times W/2}$, and then generates multiscale cross-layer feature weights through the SE attention mechanism

$$\beta = \sigma\left(W_2\delta\left(W_1 GAP\left(F_{cat}\right)\right)\right) \tag{16}$$

where $W_1 \in R^{d\times2C}$ and $W_2 \in R^{2C\times d}$ represent the weight of the fully connected layer, and δ is the ReLU activation function. AH-CEF uses this weight to drive bidirectional feature compensation

$$\left(\tilde{F}\right)_{low} = F_{low} + (\beta)_{high} \odot (Upsample)_{\times2}\left(F_{high}\right) \tag{17}$$

$$\left(\tilde{F}\right)_{high} = F_{high} + (\beta)_{low} \odot (AvgPool)_{\times2}\left(F_{low}\right) \tag{18}$$

where $\widetilde{F}_{low}$and $\widetilde{F}_{high}$ represent the compensated features, respectively, and $\odot$ represents the channel-by-channel multiplication. The compensated low-level feature $\widetilde{F}_{low}$ is then split into multiple subfeatures, and the hollow convolution with the expansion rates of 1–3 is applied, and then spliced and outputted with the compensated high-level feature $\widetilde{F}_{high}$, and retains the residual connection.

2) Attention-Guided Resolution Adaptation: To optimize the resolution transformation process, we propose an ARGA module and use it in HSFN. The ARGA module includes upsampling and downsampling branches, and designs adaptive paths for upsampling and downsampling, respectively. In the upsampling branch, ARGA uses dual paths to generate high-resolution features, retaining local structural details through deconvolution paths, and maintaining global semantic consistency using interpolation–convolution paths. The two filter key information through the channel attention gating mechanism and dynamically weighted fusion, suppressing redundant noise and enhancing key area responses. The channel attention weight is

$$G = HardSigmoid\left(W_2 \cdot \delta\left(W_1 \cdot GAP\left(F\right)\right)\right) \in ([0,1])^C \tag{19}$$

where $W_1 \in \mathbb{R}^{d\times C}$ and $W_2 \in \mathbb{R}^{C\times d}$ represent the fully connected layer weight matrix, while $d = \lfloor C/r \rfloor$, and r represents the compression ratio. $HardSigmoid(\cdot)$ is a piecewise linear activation function, and $GAP(\cdot)$ is the global average pooling. The upsample and fusion output phases can be specifically expressed as

$$F_{out} = G \odot \left[F_{path1} \oplus F_{path2}\right] \in R^{C\times2H\times2W} \tag{20}$$

where $\oplus$ represents the channel dimension splicing, F_{path1} represents the deconvolution path, and F_{path2} represents the interpolation–convolution path. In the downsampling branch, ARGA still uses dual paths to fully retain feature information and generate low-resolution features, where the stepper convolution path captures high-frequency edge information, while the pooling–convolution path retains low-frequency semantic features, and ultimately, the key features in the dual paths are dynamically screened through the channel attention gating

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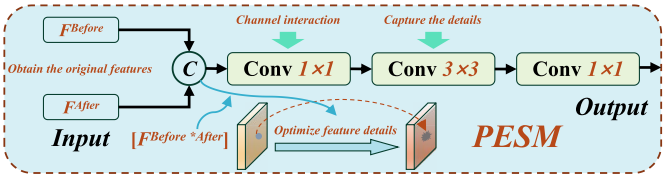


Fig. 6. Structure of PESM.

mechanism to avoid information degradation caused by resolution reduction. The downsampling fusion output phase can be specifically expressed as

$$F_{out} = G\bigodot\left[F_{path3} \oplus F_{path4}\right] \in R^{C\times H\times H} \tag{21}$$

where F_{path3} represents the stepper convolution path, and F_{path4} represents the pooled–convolution path. This process can effectively take into account global features and local details, and effectively reduce computing overhead.

3) Progressive Edge Strengthening Module: To perform secondary refinement of feature information, we propose an adaptive context edge fusion module to further fuse multiscale semantics of cross-layer features. At the same time, an attention-gated upsampling/downsampling module is used to retain key features through dual-path interpolation and channel attention screening to avoid edge blurring caused by conventional interpolation. As shown in Fig. 6, in order to improve the detailed retention ability of the model, we extracted the shallow information extracted from the backbone network twice, cleverly combined edge information and ordinary convolutional features, and proposed a new cross-channel feature fusion method, PESM. The core is to quadratically refine the extracted feature information through the three-level convolution progressive extraction mechanism. Specifically, after receiving multilevel feature input, PESM first uses channel splicing to integrate multiscale information, and then adopts a three-level convolution progressive extraction mechanism. First uses 1×1 convolution to map the spliced high-dimensional features to low-dimensional space, then performs dense spatial convolution on the dimensional reduction feature map through 3×3 convolution, and finally uses 1×1 convolution to adjust the edge-enhanced features to the target channel number, and improves feature separability through nonlinear activation functions. By combining the progressive hierarchical structure and parameter sharing strategy, this method can effectively improve the network's ability to integrate edge features in complex environments.

IV. Experimental Results

In the preliminary stage of the experiment, we define small goals, medium goals, and large goals. Specifically, we refer to the absolute scale-based object size definition established by the COCO [47] dataset for object detection and used for all experiments in this article. That is, the target with a pixel area less than 32×32 pixels is defined as a small target, the target with a pixel area between 32×32 and 96×96 is a medium-sized target, and the target with a pixel area more than 32×32 pixels is called a large target.

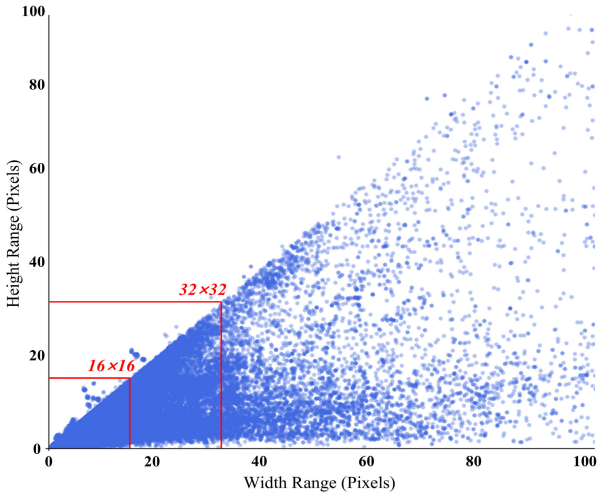


Fig. 7. Distribution of object sizes in DOTAv1.0 dataset.

A. Dataset

To verify the reliability of RSRT-DETR, we selected three public datasets in the experimental dataset section for verification. They are DOTAv1.0 [48], VisDrone2019 [49], NWPU VHR-10 [50], DIOR datasets [51], and HIT-UAV [52] datasets, respectively.

1) DOTAv1.0: As a general dataset, DOTAv1.0 contains aerial images from different sensors and platforms, with a total of 2806 remote sensing images, including 188 282 object instances. Image sizes range from 800×800 to 4000×4000. Training a model with original images is challenging due to the oversized image size. Therefore, we use long-side notation to crop and integrate the dataset into 13 554 images. We analyzed the cropped dataset, as shown in Fig. 7, in which small-size targets account for a large proportion of the total target number, which puts forward high requirements for the detection performance of the model. The processed dataset image size is 1024×1024, and the step size is 824. This dataset consists of 15 categories of objects: small vehicles (SVs), large vehicles (LVs), plane (PL), storage tanks (ST), ships (SH), harbor (HA), track and field fields (GTF), football fields (SBF), tennis courts (TC), swimming pools (SPs), baseball fields (BDs), roundabouts (RA), basketball courts (BCs), bridges (BR), and helicopters (HC). In terms of labeling, we minimize the horizontal ground reality box by minimizing the horizontal outer rectangle of the original directional ground reality box trained and tested. Therefore, 10 276 of these images were used for training, and the remaining 3278 images were used as the verification set for verification.

2) VisDrone2019: The VisDrone2019 dataset is a large-scale object detection dataset for drone aerial photography scenarios. It contains 8629 images covering different regions, different seasons, and lighting conditions, a total of 343 205 labels. As one of the most challenging object detection benchmark datasets from the perspective of drones, each image in the VisDrone2019 dataset contains an average of 53 instance objects, and most of the detection objects are small targets. All targets in the dataset are manually marked with axially aligned bounding boxes, with image sizes ranging from 960×540 to

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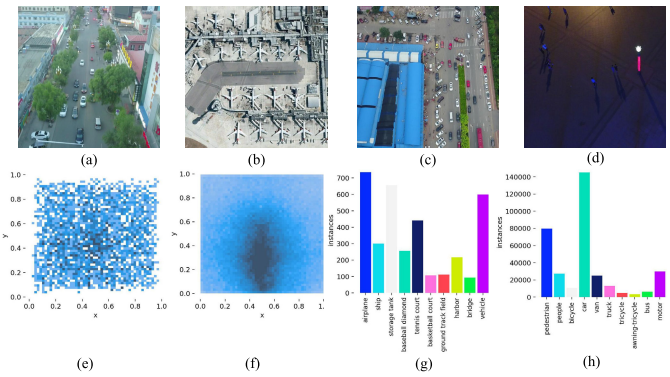


Fig. 8. Joint analysis of the VisDrone2019 and the NWPU VHR-10 datasets is shown, and images taken by drones under different conditions in the two datasets are displayed. (a) Scene in which the target is partially blocked. (b) Dense small-target scenarios in complex environments. (c) Scenes for different features of the same target. (d) Low visibility scenario. (e) Aspect ratio of the target size in the VisDrone2019 dataset. (f) Aspect ratio of the target size in the NWPU VHR-10 dataset. (g) Number of comments occupied by each category in the VisDrone2019 dataset. (h) Number of comments occupied by each category in the NWPU VHR-10 dataset.

2000 × 1500 pixels, including ten common target categories: pedestrians, crowds, bicycles, cars, vans, trucks, tricycles, sunshade tricycles, buses, and motorcycles [as shown in Fig. 8(g)]. We use the format of the official dataset to allocate weights for training, verification, and testing. That is, the training set is 6471 images, the verification set is 548 images, and the test set is 1610 images.

3) NWPU VHR-10: The NWPU VHR-10 dataset is a public reference dataset for high-resolution remote sensing image object detection. It is mainly composed of visible light and partial infrared remote sensing images obtained by satellite and aviation platforms, covering a variety of complex scenarios, such as cities, ports, airports, and farmland, including typical land objects under different scales, postures, and backgrounds. It contains 800 high-resolution remote sensing images, labeling 3651 target instances. As shown in Fig. 8(h), the image spatial resolution of the NWPU VHR-10 dataset is 0.5–2 m, covering ten categories: aircraft, ships, oil storage tanks, baseball courts, TC, BCs, track and field fields, ports, bridges, and vehicles. We use 550 of these images as the training set, 100 images as the verification set, and 150 images as the test set.

4) DIOR: The DIOR dataset comprises a total of 23 463 optical remote sensing images at a resolution of 800 × 800 pixels, with spatial resolutions ranging from 0.5 to 30 m. Among them, 5862 images are used for training, 5863 for validation, and 11 738 for testing. The training and validation sets together cover 20 common object categories with a total of 190 288 annotated instances: airplane (c1), airport (c2), BD (c3), BC (c4), bridge (c5), chimney (c6), dam (c7), expressway service area (c8), expressway toll station (c9), golf course (c10), running track (c11), harbor/port (c12), overpass/interchange (c13), ship/vessel (c14), stadium (c15), storage tank (c16), TC (c17), train station (c18), vehicle (c19), and wind turbine (c20).

5) HIT-UAV: The HIT-UAV dataset is an infrared thermal imaging dataset designed for high-altitude target detection applications using uncrewed aerial vehicles (UAVs). It comprises 2898 infrared thermal images extracted from 43 470 frames across hundreds of UAV-captured videos, with all images uniformly standardized to a resolution of 640 × 512 pixels. The dataset covers diverse real-world scenarios, including schools, parking lots, roads, and playgrounds. It provides rich flight metadata, including flight altitudes ranging from 60 to 130 m, camera angles from 30° to 90°, as well as capture dates and lighting conditions (daytime/nighttime). The data is divided into three standard subsets: 2029 images for training, 579 for testing, and 290 for validation. The entire dataset contains annotations for 24 899 object instances, averaging 8.59 instances per image—significantly higher than many general-purpose datasets. HIT-UAV focuses on five common urban object categories: person, car, bicycle, other vehicle, and an "uncertain" category (DontCare) for unclassifiable objects. Notably, to address the challenge of severe object overlap in aerial imagery, each instance in the dataset provides two types of bounding box annotations: standard horizontal rectangles and more precise oriented rectangles. HIT-UAV is the first publicly available high-altitude drone infrared thermal imaging dataset tailored for person and vehicle detection. It aims to advance research and development in drone applications such as nighttime surveillance, traffic management, and search and rescue operations.

B. Settings

All experiments were conducted on a workstation equipped with an Intel Core i7-13900K CPU and a single NVIDIA GeForce RTX 4090 GPU, running on the Ubuntu 20.04 operating system. The implementation is based on the PyTorch 2.1.0 deep learning framework with Python 3.10 and CUDA 12.2. RSRT-DETR utilizes ResNet [53]-50 or ResNet-101 pre-trained on ImageNet [54] as the backbone network to extract multiscale features. The model was trained using the AdamW optimizer with an initial learning rate of $\mathbf{1} \times \mathbf{10^{-4}}$, a weight decay of $\mathbf{1} \times \mathbf{10^{-4}}$, and a momentum of 0.937. The batch size was set to 4. To ensure a strictly fair and robust comparison, we standardized the backbone networks of all Transformer-based variants to ResNet-50 and ResNet-101. For the YOLO series and other state-of-the-art baselines, we strictly adhered to their official open-source implementations and optimal hyperparameter configurations. Specifically, the YOLO comparison models were trained for 200 epochs employing their native aggressive data augmentation strategies to guarantee their peak performance. In contrast to rigorously evaluating the efficacy of our proposed architectural modules rather than relying on training tricks, RSRT-DETR was trained for only 100 epochs using standard data augmentations. These settings effectively demonstrate that RSRT-DETR exhibits stronger detection performance compared to other models and is more suitable for remote sensing image target detection tasks.

C. Evaluation Metrics

In order to comprehensively evaluate the effectiveness of RSRT-DETR, this study adopted mean accuracy (AP) as

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Fig. 9. Partial results after detection using RSRT-DETR on DOTAv1.0 dataset.

the main evaluation indicator. To understand the concept of AP in depth, we introduced two basic indicators of recall and precision, which represent the proportion of correctly identified positive samples and the proportion of correctly identified positive instances among all instances identified as positive. Specifically, expressed as follows:

$$Recall = \frac{TP}{TP + FN} \tag{22}$$

$$Precision = \frac{TP}{TP + FP}. \tag{23}$$

Among them, true positives (TP) represent the number of bounding boxes correctly detected, false positives (FP) represent the number of bounding boxes incorrectly detected, and false negatives (FN) represent the number of bounding boxes missing during the detection process of the model. Furthermore, average accuracy (AP) represents the tradeoff between accuracy and accuracy, which is quantified as the area under the accuracy-view curve. The calculation formula of AP is as follows:

$$AP = \frac{1}{m}\sum_{i}^{m} P_i = \int P(R)\, dR. \tag{24}$$

It is worth noting that all instances of AP in this article specifically involve AP_{50}. On this basis, we introduced small-objective accuracy (AP_S) and medium-objective detection accuracy (AP_M) to illustrate that RSRT-DETR has higher sensitivity to small-objective models. At the same time, we apply Parameters (number of model parameters) and float per second (FLOPs) to evaluate the size and computational efficiency of the model. In addition, frames per second (FPS) is used to measure the model's inference speed.

D. Comparative Experiments With Mainstream Methods and SOTA Models

In order to better prove the excellent performance of RSRT-DETR, we have introduced various mainstream methods currently used to compare with the SOTA model, including but not limited to the CNN-based object detection model and the Transformer-based object detection model. The CNN-based object detection model includes the second-stage model Faster-RCNN [14], the first-stage detection model SSD [15] and YOLOv11 [55], and the Transformer-based object detection model includes PSWP-DETR [43], OWRT-DETR [56], SED-DETR [38], and RT-DETR [46] and many other

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Fig. 10. Examples of detection results on the Visdrone2019 dataset using RSRT-DETR.



Fig. 11. Partial results of detection using RSRT-DETR on NWPU VHR-10 dataset.

DETR-based SOTA models. Figs. 9–11 show the detection results of RSRT-DETR and the remaining models on DOTAv1.0, Visdrone2019, and NWPU VHR-10 datasets. It can be seen that RSRT-DETR has achieved the optimal detection capability of dense small targets on three datasets, while other models have different degrees of missed and false detection in these three datasets. Thanks to the unique design of RSRT-DETR, it has achieved excellent performance in the detection of dense small targets in complex contexts.

1) Comparison on DOTAv1.0 Dataset: Table I and Fig. 12 show the quantitative comparison results of RSRT-DETR on the DOTAv1.0 dataset. We can find that RSRT-DETR achieves the highest detection accuracy compared to other mainstream methods that also use the ResNet 50 backbone, compared to the SOTA model. Compared with the CNN-based object detection model, because the Transformer has higher global modeling capabilities, RSRT-DETR achieves higher detection accuracy compared with it. Thanks to the high sensitivity of PDSF Attention and HSFN to small targets, RSRT-DETR also achieved the best detection accuracy of small and medium targets. At the same time, MS-HFNet effectively reduces the model's parameter quantity, so that although the model is not the lightest in terms of FLOPs and parameter quantity, it also achieves relatively good results and is suitable for drone detection tasks. Compared with the latest SOTA models, such as PSWP-DETR, OWRT-DETR, and SED-DETR, RSRT-

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TABLE I
COMPARISON OF OBJECT DETECTION METHODS ON DOTAV1.0 DATASET

Model	Backbone	Epochs	Params (M)	FLOPs (G)	AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based									
Faster-RCNN [14]	R50	24	52.1	237.2	17.3	17.8	47.6	27.5	22.3
SSD [15]	R50	24	32.7	89.3	12.4	13.6	46.5	26.3	21.6
CenterNet [57]	R101	100	48.3	112.6	13.5	13.9	45.2	25.1	20.8
EfficientDet [32]	-	100	32.7	593.4	14.8	13.7	46.3	25.9	21.2
YOLOv5-L [58]	-	200	31.6	186.9	19.8	18.4	56.2	29.1	27.8
YOLOv5-X [58]	-	200	63.7	226.8	20.2	19.1	56.3	30.8	28.6
YOLOv8-S [59]	-	200	15.7	24.8	23.9	19.9	62.7	34.1	32.0
YOLOv8-M [59]	-	200	24.8	72.8	21.4	21.9	63.9	32.2	32.8
YOLOv8-L [59]	-	200	29.7	172.1	23.0	19.6	61.8	33.6	31.4
YOLOv8-X [59]	-	200	59.5	216.2	20.5	20.3	60.3	32.3	32.1
YOLOv9-S [60]	-	200	28.1	108.4	21.1	19.9	62.6	32.9	32.9
YOLOv9-M [60]	-	200	32.3	123.5	20.9	18.9	63.3	32.3	33.5
YOLOv9-L [60]	-	200	39.2	134.1	21.5	19.3	63.6	33.7	34.5
YOLOv9-X [60]	-	200	63.5	198.2	21.9	19.7	64.2	33.2	33.6
YOLOv10-S [61]	-	200	28.9	126.7	18.6	16.4	65.3	31.6	30.7
YOLOv10-M [61]	-	200	31.6	103.5	19.6	17.9	66.3	32.5	30.7
YOLOv10-L [61]	-	200	42.3	136.3	20.2	17.4	66.6	33.8	31.0
YOLOv10-X [61]	-	200	60.7	197.6	20.7	17.9	66.2	33.2	32.0
YOLOv11-S [55]	-	200	10.3	21.9	21.0	18.2	69.2	33.7	40.6
YOLOv11-M [55]	-	200	22.6	70.3	20.5	18.5	69.4	32.9	39.8
YOLOv11-L [55]	-	200	26.1	89.3	21.3	18.9	68.1	32.3	32.6
YOLOv11-X [55]	-	200	57.2	197.5	20.8	18.3	68.4	32.1	32.9
YOLOv12-S [62]	-	200	10.1	20.6	22.3	18.3	67.5	31.9	42.6
YOLOv12-M [62]	-	200	20.6	59.5	21.4	18.9	68.3	33.5	40.7
YOLOv12-L [62]	-	200	28.4	89.3	21.2	17.4	68.5	34.8	39.7
YOLOv12-X [62]	-	200	59.6	199.7	21.5	17.9	69.2	34.2	40.6
Transformer Based									
Anchor-DETR [40]	R50	50	38.4	176.5	25.7	22.3	60.8	31.3	32.9
Anchor-DETR [40]	R101	50	56.7	231.1	22.3	19.7	53.7	30.6	33.0
Conditional-DETR [63]	R50	50	45.6	67.2	11.6	10.3	50.9	19.7	24.5
Conditional-DETR [63]	R101	50	66.8	229.2	15.6	13.5	51.3	23.4	26.4
Deformable-DETR [37]	R50	50	42.8	173.6	20.7	18.6	58.6	28.3	31.5
Deformable-DETR [37]	R101	50	60.2	228.2	25.7	22.4	62.4	29.5	32.7
DAB-DETR [39]	R50	50	51.6	165.8	34.7	30.8	67.9	40.8	40.6
DAB-DETR [39]	R101	50	72.4	228.7	35.4	30.2	66.3	42.9	41.5
DN-DETR [41]	R50	50	49.2	163.3	34.7	29.8	67.5	41.3	40.9
DN-DETR [41]	R101	50	70.3	224.5	34.3	30.7	65.1	43.1	40.8
Focus-DETR [42]	R50	36	48.2	112.5	35.1	30.2	67.3	41.7	41.1
Focus-DETR [42]	R101	36	67.5	224.2	36.2	31.8	67.9	42.4	41.4
DINO-4scale [34]	R50	12	47.5	179.8	32.9	28.4	66.2	39.3	39.6
DINO-4scale [34]	R101	12	68.1	267.4	36.5	31.3	67.5	39.6	40.0
SC-RTDETR [44]	R50	50	39.2	140.6	33.5	29.1	63.8	37.6	38.0
SC-RTDETR [44]	R101	50	57.8	245.6	34.9	30.5	66.2	40.1	40.5
ARS-DETR [45]	R50	36	43.7	163.8	32.8	28.3	64.5	38.2	38.9
ARS-DETR [45]	R101	36	65.2	241.7	34.1	29.8	66.3	40.5	40.9
PSWP-DETR [43]	R50	72	98.8	155.0	33.2	28.3	64.0	38.5	39.6
PSWP-DETR [43]	R101	72	141.6	235.9	34.5	29.9	66.5	40.8	41.2
OWRT-DETR [56]	R50	100	40.6	126.4	31.3	29.9	63.7	40.5	40.9
OWRT-DETR [56]	R101	100	64.5	223.3	34.5	31.2	65.6	40.6	41.3
GMODet [64]	R50	200	44.5	168.0	33.2	28.7	64.2	38.9	39.3
GMODet [64]	R101	200	143.9	242.3	34.8	30.3	66.8	41.2	41.6
SED-DETR [38]	R50	50	48.4	61.9	34.3	28.1	64.3	38.6	39.7
SED-DETR [38]	R101	50	52.1	73.6	34.9	28.7	64.6	39.1	40.2
RT-DETR [46]	R50	72	41.6	134.4	34.7	30.2	64.4	39.2	39.8
RT-DETR [46]	R101	72	68.9	256.4	35.8	31.2	67.3	41.7	41.1
RSRT-DETR (Ours)	R50	100	36.7	105.1	36.3	30.6	69.3	40.9	41.1
RSRT-DETR (Ours)	R101	100	64.9	200.8	36.8	31.2	69.7	41.3	41.9

DETR has achieved an average improvement rate of 3.5% on each AP indicator. In addition, RSRT-DETR still achieves optimal results in multiple indicators compared with other mainstream methods that also use the ResNet 101 backbone,

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TABLE II
COMPARISON OF OBJECT DETECTION METHODS ON VISDRONE2019 DATASET

Model	Backbone	Epochs	Params (M)	FLOPs (G)	AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based									
Faster-RCNN [14]	R50	24	56.9	241.3	8.3	18.2	42.1	12.7	20.5
SSD [15]	R50	24	35.4	95.2	6.1	16.7	38.9	10.2	18.3
CenterNet [57]	R101	100	50.5	117.8	7.9	19.1	44.3	14.6	22.3
EfficientDet [32]	-	100	37.4	597.6	5.7	15.9	36.8	9.8	17.1
YOLOv5-L [58]	-	200	35.8	191.2	11.6	24.6	48.7	18.3	26.5
YOLOv5-X [58]	-	200	66.3	232.7	12.3	25.3	50.2	20.1	28.3
YOLOv8-L [59]	-	200	33.8	176.4	13.7	29.8	54.3	23.6	31.2
YOLOv8-X [59]	-	200	64.7	221.5	14.9	31.2	56.1	25.4	33.1
YOLOv8-S [59]	-	200	19.6	29.5	12.6	22.7	45.9	16.2	24.3
YOLOv8-M [59]	-	200	30.3	77.1	12.1	26.9	51.3	20.8	28.7
YOLOv9-S [60]	-	200	30.7	112.9	13.2	28.4	52.7	22.3	30.1
YOLOv9-M [60]	-	200	37.2	128.1	14.6	30.1	54.6	24.1	32.6
YOLOv9-L [60]	-	200	43.1	138.6	15.8	31.7	56.2	26.4	33.8
YOLOv9-X [60]	-	200	68.2	203.9	16.9	33.2	56.7	27.6	35.4
YOLOv10-S [61]	-	200	33.5	131.6	14.1	29.3	53.8	23.7	31.2
YOLOv10-M [61]	-	200	36.3	109.2	15.3	30.8	55.4	25.3	33.2
YOLOv10-L [61]	-	200	46.8	141.9	16.7	32.5	57.1	27.1	35.1
YOLOv10-X [61]	-	200	65.2	202.3	17.9	34.1	58.6	28.8	36.7
YOLOv11-L [55]	-	200	23.5	85.4	14.8	30.5	55.1	24.9	32.6
YOLOv11-X [55]	-	200	52.9	191.3	16.2	32.3	57.2	27.4	34.9
YOLOv11-S [55]	-	200	9.9	22.8	16.8	27.5	51.9	25.1	30.4
YOLOv11-M [55]	-	200	18.2	66.3	17.6	28.3	54.8	27.3	34.3
YOLOv12-S [62]	-	200	9.1	19.6	12.9	27.5	52.8	21.4	29.1
YOLOv12-M [62]	-	200	19.6	60.1	14.8	30.2	55.1	24.3	32.4
YOLOv12-L [62]	-	200	25.8	81.7	16.1	32.6	56.3	26.5	34.5
YOLOv12-X [62]	-	200	57.9	182.6	17.5	33.7	57.9	28.4	36.3
Transformer Based									
Anchor-DETR [40]	R50	50	43.9	181.8	12.2	25.3	49.1	18.9	27.2
Anchor-DETR [40]	R101	50	61.3	236.7	12.5	27.1	51.8	21.9	29.1
Conditional-DETR [63]	R50	50	50.3	72.9	11.8	24.7	48.3	18.1	26.5
Conditional-DETR [63]	R101	50	71.5	234.8	12.1	26.8	51.2	20.6	28.7
Deformable-DETR [37]	R50	50	47.3	178.2	13.6	28.9	53.2	22.8	30.7
Deformable-DETR [37]	R101	50	65.7	233.8	14.9	30.7	55.1	24.9	32.8
DAB-DETR [39]	R50	50	56.3	171.5	12.3	26.9	50.7	20.3	28.6
DAB-DETR [39]	R101	50	77.3	233.3	13.7	28.8	52.9	22.5	30.7
DN-DETR [41]	R50	50	54.7	168.4	14.1	29.3	53.6	23.1	31.3
DN-DETR [41]	R101	50	75.9	229.2	15.4	31.1	55.4	25.2	33.2
Focus-DETR [42]	R50	36	53.8	117.2	16.8	30.4	55.3	24.6	32.1
Focus-DETR [42]	R101	36	72.1	229.7	16.1	32.1	56.0	26.2	34.3
DINO-4scale [34]	R50	12	52.1	185.5	16.6	31.5	56.9	27.8	34.9
DINO-4scale [34]	R101	12	73.7	272.1	17.4	33.3	57.3	29.6	35.5
SC-RTDETR [44]	R50	50	44.3	145.3	15.3	29.9	53.8	23.6	31.5
SC-RTDETR [44]	R101	50	62.4	251.3	15.7	31.8	55.9	25.9	33.8
ARS-DETR [45]	R50	50	48.2	168.5	14.9	29.1	52.9	22.7	30.6
ARS-DETR [45]	R101	50	70.8	247.3	15.3	31.3	55.2	25.1	33.1
PSWP-DETR [43]	R50	50	103.5	160.7	15.7	30.6	54.4	24.3	32.4
PSWP-DETR [43]	R101	50	146.3	241.6	16.9	32.4	56.3	26.5	34.5
OWRT-DETR [56]	R50	100	45.3	131.1	15.1	30.9	54.8	24.8	32.9
OWRT-DETR [56]	R101	100	69.2	228.1	16.4	32.8	56.9	26.9	35.7
GMODet [64]	R50	50	49.2	173.7	13.2	28.4	51.8	21.9	29.8
GMODet [64]	R101	50	148.6	247.3	14.6	30.4	53.9	24.1	32.5
SED-DETR [38]	R50	50	53.4	67.6	12.7	27.3	51.6	21.6	28.7
SED-DETR [38]	R101	50	57.8	79.3	13.9	29.1	52.4	22.5	30.5
RT-DETR [46]	R50	72	46.3	139.1	16.2	29.3	54.2	31.7	34.3
RT-DETR [46]	R101	72	74.6	261.1	16.8	32.6	56.8	27.4	35.1
RSRT-DETR (Ours)	R50	100	41.4	110.8	17.8	31.7	56.3	32.9	35.6
RSRT-DETR (Ours)	R101	100	69.5	206.4	17.2	32.9	57.1	28.6	35.4

compared with the SOTA model, significantly demonstrating its powerful performance on target detection in complex scenarios.

2) Comparison on VisDrone2019 Dataset: To verify the generalization ability of RSRT-DETR, we also conducted experiments on VisDrone2019. As shown in Table II and

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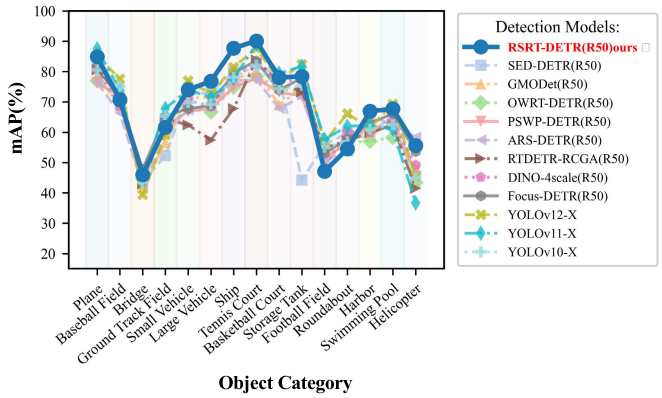


Fig. 12. Visualize the comparison results of RSRT-DETR and many mainstream models on each category on the DOTAv1.0 dataset.

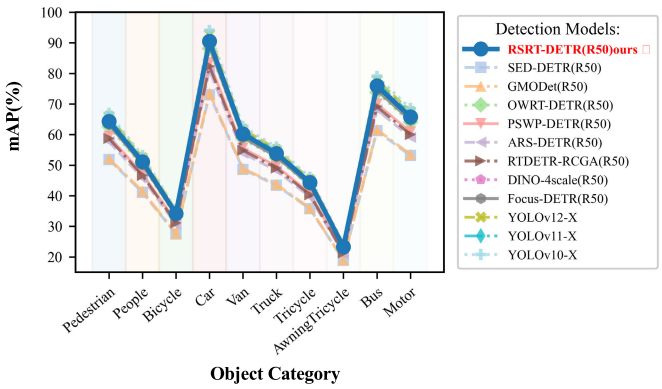


Fig. 13. Visualize the comparison results of RSRT-DETR and many mainstream models on each category on the Visdrone2019 dataset.

Fig. 13, RSRT-DETR achieved AP_{50} of up to 56.3% on the ResNet 50 backbone, an increase of 0.4%–7.2% compared with the remaining Transformer-based object detection models. Similarly, on the ResNet 101 backbone, RSRT-DETR achieved an AP_{50} of up to 57.1%, which increased by 0.2% to 5.9% compared with the remaining Transformer-based object detection models. It is worth noting that RSRT-DETR achieves optimal performance in AP_S indicators for the detection of small targets, a progress that demonstrates the excellent performance of RSRT-DETR to detect dense small targets in complex contexts. In Fig. 10, we can see that RSRT-DETR achieves accurate identification of most targets and demonstrates its excellent performance.

3) Comparison on NWPU VHR-10 Dataset: In this subsection, we put RSRT-DETR together with numerous other models into the NWPU VHR-10 dataset for testing. As a dataset covering multiple challenges such as large-scale differences, uneven category distribution, and complex backgrounds, the NWPU VHR-10 dataset puts forward high requirements for the model's fine-grained feature extraction and multiscale detection capabilities. Table III and Fig. 14 show the experimental results of RSRT-DETR and other models in the NWPU VHR-10 dataset. Also, Fig. 11 shows the inference results of RSRT-DETR on this dataset. It can be seen that RSRT-DETR still achieves the highest detection accuracy compared with other CNN-based object detection

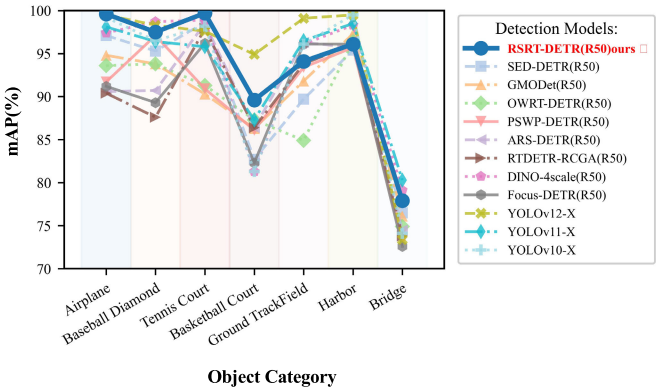


Fig. 14. Visualize the comparison results of RSRT-DETR and many mainstream models on each category on the NWPU VHR-10 dataset.

models and the Transformer object detection models with the same benchmark model, and has improved by 0.7%–21.1% on the AP_{50} indicator. At the same time, we can still find from the AP_S indicator that RSRT-DETR still achieves the highest accuracy, which shows that the RSRT-DETR algorithm detects dense small targets more accurately and comprehensively. In addition, according to the comprehensive results of the three datasets, RSRT-DETR still has extremely high detection capabilities in different environments and different targets, and at the same time, it has achieved a good balance in detection accuracy and computing overhead. It can be used to achieve efficient and high-precision identification and positioning of dense small targets in complex contexts.

4) Comparison on DIOR Dataset: As shown in Table IV, RSRT-DETR delivers state-of-the-art performance among Transformer-based detectors on the DIOR dataset while remaining competitive with leading CNN-based models. Using the R50 backbone, RSRT-DETR attains AP_{50} = 88.1 and $AP_{50:95}$ = 66.8 with 41.3M params and 111.5 GFLOPs; the R101 variant further lifts accuracy to AP_{50} = 89.8 and $AP_{50:95}$ = 67.8 at 73.8M/208.2 GFLOPs. Notably, RSRT-DETR achieves the best small-object accuracy across all methods, with AP_S of 77.1 (R50) and 78.8 (R101), surpassing recent CNN baselines, such as YOLOv12-X (77.6) and YOLOv11-X (72.6). Compared with representative Transformer baselines (e.g., Deformable-DETR, DAB-DETR, DN-DETR, DINO-4scale, RT-DETR, and variants), RSRT-DETR (R101) achieves higher AP_{50} and $AP_{50:95}$ while maintaining moderate complexity. These results indicate that RSRT-DETR provides more accurate and robust detection—especially on dense small targets—striking a favorable balance between detection quality and computational cost on the DIOR dataset.

5) Comparison on HIT-UAV Dataset: In comparative experiments on the HIT-UAV drone target detection dataset, various models demonstrated significant performance differences (Table V). Among them, the YOLO series models stood out notably, achieving an AP50 score as high as 85.2%. The RSRT-DETR model, based on the Transformer architecture and utilizing R50 and R101 backbone networks, achieved AP50 results of 83.9% and 82.3%, respectively. This represents a significant improvement over comparable Transformer

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TABLE III
COMPARISON OF OBJECT DETECTION METHODS ON NWPU VHR-10 DATASET

Model	Backbone	Epochs	Params (M)	FLOPs (G)	AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based									
Faster-RCNN [14]	R50	24	51.5	207.3	36.9	58.1	74.3	40.2	42.8
SSD [15]	R50	24	37.6	91.3	31.2	50.6	68.9	35.1	38.5
CenterNet [57]	R101	100	63.4	285.7	41.8	62.3	79.1	45.6	47.2
EfficientDet [32]	-	100	33.7	585.4	39.5	59.2	76.8	42.1	44.3
YOLOv5-L [58]	-	200	46.5	109.1	48.7	66.5	84.2	53.2	55.9
YOLOv5-X [58]	-	200	86.7	205.8	50.1	68.3	85.6	55.7	57.8
YOLOv8-S [59]	-	200	11.1	28.5	83.3	76.2	91.5	52.7	60.1
YOLOv8-M [59]	-	200	25.8	78.7	86.9	78.4	92.3	54.2	61.4
YOLOv8-L [59]	-	200	43.6	164.9	83.3	75.1	91.3	52.9	64.6
YOLOv8-X [59]	-	200	68.1	257.4	82.2	77.3	90.9	53.4	62.7
YOLOv9-S [60]	-	200	11.8	32.1	84.5	76.8	91.7	53.8	61.0
YOLOv9-M [60]	-	200	21.6	85.6	85.9	78.1	92.5	55.1	62.5
YOLOv9-L [60]	-	200	32.4	178.3	87.2	80.2	93.2	56.8	63.9
YOLOv9-X [60]	-	200	56.3	299.4	88.1	79.4	93.8	57.5	65.2
YOLOv10-S [61]	-	200	7.2	21.4	83.3	73.1	89.7	52.9	58.1
YOLOv10-M [61]	-	200	15.3	58.9	84.7	72.2	89.6	53.6	60.9
YOLOv10-L [61]	-	200	24.3	120.0	83.3	71.3	91.5	55.5	62.0
YOLOv10-X [61]	-	200	29.4	160.0	84.8	78.5	92.1	57.2	59.8
YOLOv11-S [55]	-	200	9.1	21.0	85.1	75.6	89.5	56.3	59.6
YOLOv11-M [55]	-	200	18.2	65.8	86.9	77.2	92.0	58.3	61.8
YOLOv11-L [55]	-	200	23.5	84.7	85.4	75.1	92.9	56.5	60.1
YOLOv11-X [55]	-	200	52.8	190.2	83.2	73.6	94.3	54.4	62.5
YOLOv12-S [62]	-	200	9.0	19.3	87.8	73.9	92.3	57.1	59.6
YOLOv12-M [62]	-	200	19.5	59.5	87.3	77.6	94.6	58.2	61.3
YOLOv12-L [62]	-	200	25.7	80.8	88.7	76.1	94.5	55.3	64.6
YOLOv12-X [62]	-	200	57.8	181.3	89.9	72.7	94.6	59.4	65.0
Transformer Based									
Anchor-DETR [40]	R50	50	40.2	179.5	70.1	62.3	80.6	43.3	48.6
Anchor-DETR [40]	R101	50	75.8	230.2	66.5	59.8	76.8	43.8	49.2
Conditional-DETR [63]	R50	50	41.5	68.2	68.7	58.9	77.9	31.7	35.6
Conditional-DETR [63]	R101	50	77.1	227.2	72.4	62.1	81.6	34.1	38.1
Deformable-DETR [37]	R50	50	34.6	174.5	74.5	65.2	81.9	46.0	51.7
Deformable-DETR [37]	R101	50	68.9	225.1	78.3	68.7	87.5	44.2	49.6
DAB-DETR [39]	R50	50	42.8	164.7	82.3	72.6	90.6	52.3	58.8
DAB-DETR [39]	R101	50	79.5	228.7	83.1	73.8	91.3	54.4	61.1
DN-DETR [41]	R50	50	43.1	163.3	84.6	74.2	93.1	53.6	60.2
DN-DETR [41]	R101	50	80.2	226.5	85.0	74.8	93.7	53.4	60.0
Focus-DETR [42]	R50	36	38.9	113.7	80.2	70.8	89.1	53.9	60.5
Focus-DETR [42]	R101	36	72.3	223.1	79.6	70.2	88.6	54.3	60.9
DINO-4scale [34]	R50	12	47.5	179.8	84.1	73.5	92.8	52.1	58.3
DINO-4scale [34]	R101	12	89.7	275.6	86.3	75.2	94.7	52.4	58.8
SC-RTDETR [44]	R50	50	44.8	142.3	81.2	70.5	89.4	50.2	55.9
SC-RTDETR [44]	R101	50	82.1	275.4	85.7	74.1	92.8	53.1	59.6
ARS-DETR [45]	R50	50	39.5	125.5	82.9	72.1	90.4	51.0	57.2
ARS-DETR [45]	R101	50	74.8	246.9	86.4	73.8	93.1	53.7	60.2
PSWP-DETR [43]	R50	50	98.9	154.1	82.3	70.3	89.7	52.0	58.3
PSWP-DETR [43]	R101	50	140.7	236.0	87.2	71.9	93.3	54.1	60.6
OWRT-DETR [56]	R50	100	41.3	124.3	83.2	75.3	88.9	51.3	56.6
OWRT-DETR [56]	R101	100	66.6	225.6	86.7	75.1	91.7	54.6	58.7
GMODet [64]	R50	50	43.5	167.3	80.9	72.3	90.0	51.5	57.8
GMODet [64]	R101	50	144.8	246.9	84.6	73.1	93.7	54.6	61.2
SED-DETR [38]	R50	50	49.7	62.5	82.4	74.4	91.1	52.6	56.7
SED-DETR [38]	R101	50	54.3	71.6	81.6	73.3	92.8	57.3	60.3
RT-DETR [46]	R50	72	43.4	134.8	80.6	71.3	92.3	53.2	59.6
RT-DETR [46]	R101	72	76.6	257.7	82.1	71.6	91.4	54.2	61.1
RSRT-DETR (Ours)	R50	100	37.7	110.4	83.9	72.9	93.5	55.6	59.7
RSRT-DETR (Ours)	R101	100	65.4	201.1	84.6	72.3	95.4	55.7	61.6

models and outperforms current mainstream SOTA models. Simultaneously, the model demonstrated relatively balanced performance in comprehensive comparisons with YOLO series models, reflecting its favorable efficiency-balancing characteristics. Overall, the proposed model demonstrates strong performance in nighttime UAV detection tasks, effectively adapting to multimodal detection scenarios. It provides a viable direction for further exploring the synergistic optimization of detection accuracy and computational efficiency.

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TABLE IV
COMPARISON OF OBJECT DETECTION METHODS ON DIOR DATASET

Model	Backbone	Epochs	Params (M)	FLOPs (G)	AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based									
Faster-RCNN [14]	R50	24	41.5	218.6	46.9	62.6	72.1	50.7	56.5
SSD [15]	R50	24	24.3	98.5	42.8	58.6	61.3	48.1	52.1
CenterNet [57]	R101	100	53.9	123.6	47.3	61.3	68.4	40.6	49.6
EfficientDet [32]	-	100	34.8	593.7	48.5	62.1	73.5	49.3	54.2
YOLOv5-L [58]	-	200	40.9	103.2	53.4	63.5	75.8	47.1	58.3
YOLOv5-X [58]	-	200	67.8	196.8	55.6	66.8	76.7	48.9	57.4
YOLOv8-L [59]	-	200	47.2	172.3	68.9	59.2	82.1	48.2	56.3
YOLOv8-X [59]	-	200	66.8	249.1	67.2	63.4	80.6	50.3	59.2
YOLOv8-S [59]	-	200	21.7	32.7	66.9	65.5	83.4	42.8	59.0
YOLOv8-M [59]	-	200	34.8	86.2	65.1	65.9	85.8	45.9	61.2
YOLOv9-S [60]	-	200	6.3	22.8	68.7	67.4	87.3	42.2	66.8
YOLOv9-M [60]	-	200	16.8	61.6	69.3	68.2	88.7	45.3	68.0
YOLOv9-L [60]	-	200	29.5	169.4	70.0	67.3	86.9	47.8	62.5
YOLOv9-X [60]	-	200	59.2	284.1	68.8	69.1	85.2	45.2	67.3
YOLOv10-S [61]	-	200	8.9	24.9	68.2	66.3	88.0	48.6	66.5
YOLOv10-M [61]	-	200	14.2	56.2	71.6	67.9	89.1	47.2	67.0
YOLOv10-L [61]	-	200	25.7	127.4	73.5	68.3	87.6	49.7	66.4
YOLOv10-X [61]	-	200	31.6	171.2	72.2	69.2	88.2	48.3	67.2
YOLOv11-L [55]	-	200	25.3	87.4	69.4	69.2	89.7	50.2	69.2
YOLOv11-X [55]	-	200	56.8	195.6	71.3	70.4	90.5	51.6	69.8
YOLOv11-S [55]	-	200	10.5	21.6	72.6	70.3	88.0	48.5	67.4
YOLOv11-M [55]	-	200	20.0	68.3	73.9	71.6	89.0	49.2	68.8
YOLOv12-S [62]	-	200	9.6	19.6	75.3	71.2	87.0	49.9	66.0
YOLOv12-M [62]	-	200	19.6	60.2	75.9	71.0	87.8	51.1	65.2
YOLOv12-L [62]	-	200	25.8	81.7	72.5	69.5	88.5	53.4	66.0
YOLOv12-X [62]	-	200	57.9	182.7	77.6	70.3	89.2	52.6	66.5
Transformer Based									
Anchor-DETR [40]	R50	50	42.9	196.6	67.9	56.8	72.8	50.2	49.6
Anchor-DETR [40]	R101	50	76.3	230.3	69.6	57.4	74.0	51.7	51.7
Conditional-DETR [63]	R50	50	39.2	79.2	68.3	54.5	75.3	52.8	51.5
Conditional-DETR [63]	R101	50	57.5	284.9	70.2	56.1	78.6	52.1	53.7
Deformable-DETR [37]	R50	50	53.6	189.7	71.5	58.5	71.9	54.3	56.0
Deformable-DETR [37]	R101	50	72.6	264.3	73.5	61.4	72.5	53.6	50.6
DAB-DETR [39]	R50	50	59.2	196.2	69.1	55.3	76.2	55.8	54.8
DAB-DETR [39]	R101	50	78.3	259.3	70.8	57.4	77.6	48.2	56.9
DN-DETR [41]	R50	50	59.5	201.6	71.9	59.3	80.4	51.3	58.8
DN-DETR [41]	R101	50	78.3	286.7	73.7	61.7	81.6	47.5	57.9
Focus-DETR [42]	R50	36	58.6	201.5	72.6	60.6	82.1	50.7	54.2
Focus-DETR [42]	R101	36	77.1	276.4	74.5	61.2	82.3	53.2	54.9
DINO-4scale [34]	R50	12	60.7	187.3	74.9	64.2	84.1	50.6	60.5
DINO-4scale [34]	R101	12	80.5	214.7	76.6	63.1	85.7	54.1	60.2
SC-RTDETR [44]	R50	50	59.3	162.4	73.3	65.2	83.8	52.6	61.8
SC-RTDETR [44]	R101	50	79.6	188.3	75.1	64.1	84.2	55.1	62.3
ARS-DETR [45]	R50	50	56.6	139.7	73.7	63.5	79.8	53.7	59.6
ARS-DETR [45]	R101	50	76.2	167.4	75.6	66.7	81.2	55.7	60.4
PSWP-DETR [43]	R50	50	97.5	177.5	74.9	62.1	86.1	55.9	60.3
PSWP-DETR [43]	R101	50	146.9	254.1	76.2	64.9	86.7	56.5	61.2
OWRT-DETR [56]	R50	100	40.6	155.2	75.9	65.1	88.4	53.9	65.9
OWRT-DETR [56]	R101	100	66.3	269.6	77.6	63.7	89.1	54.6	65.1
GMODet [64]	R50	50	45.9	172.6	74.8	63.4	84.3	54.2	66.3
GMODet [64]	R101	50	103.6	287.1	76.3	65.8	85.9	54.7	67.1
SED-DETR [38]	R50	50	49.7	69.2	77.2	65.3	86.2	55.2	63.8
SED-DETR [38]	R101	50	54.3	80.3	72.8	66.7	87.4	55.7	64.5
RT-DETR [46]	R50	72	42.9	134.8	71.4	64.1	85.4	55.6	62.7
RT-DETR [46]	R101	72	76.6	257.7	73.5	66.4	86.4	54.9	63.2
RSRT-DETR (Ours)	R50	100	41.3	111.5	77.1	66.8	88.1	56.2	66.8
RSRT-DETR (Ours)	R101	100	73.8	208.2	78.8	68.2	89.8	55.8	67.8

E. Ablation Experiment

To analyze the importance of each component in RDRT-DETR, we took ResNet 50 as the backbone network and gradually applied different combinations of PDSF Attention, MS-HFNet, and HSFN in the baseline model to verify their effectiveness. The ablation experiment was performed on the DOTAv1.0 dataset. As shown in Table VI, where "✔" means that the module has been used, and "✘" means that the module is not used.

1) PDSF Attention: From Table VI, we can see that the performance of the model has been significantly improved after using PDSF Attention. Specifically, AP_{50} has been increased from 64.4% to 66.9% (a gain of 2.5%), while the detection accuracy of small targets (AP_S) has been increased

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TABLE V
COMPARISON OF OBJECT DETECTION METHODS ON HIT-UAV DATASET

Model	Backbone	Epochs	Params (M)	GFLOPs	AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based									
Faster-RCNN [14]	R50	24	45.9	216.7	22.5	45.1	72.4	40.1	38.5
SSD [15]	R50	24	24.8	88.2	15.3	38.4	63.8	32.5	30.2
CenterNet [57]	R101	100	53.5	143.8	26.9	48.7	69.4	42.3	41.6
EfficientDet [32]	-	100	39.7	579.4	30.1	52.4	77.5	51.2	50.1
YOLOv5-L [58]	-	200	46.5	109.1	33.5	54.2	80.5	55.1	51.2
YOLOv5-X [58]	-	200	77.9	205.3	34.6	55.6	81.2	56.5	50.6
YOLOv8-L [59]	-	200	43.2	165.1	36.5	55.1	83.5	56.2	51.1
YOLOv8-X [59]	-	200	68.5	257.9	37.1	60.5	83.9	59.8	52.9
YOLOv8-S [59]	-	200	19.1	28.7	36.1	55.8	80.1	61.2	53.8
YOLOv8-M [59]	-	200	25.8	78.6	36.8	57.4	82.4	62.5	53.7
YOLOv9-S [60]	-	200	7.2	26.7	35.5	55.8	80.3	56.8	51.4
YOLOv9-M [60]	-	200	19.9	76.4	37.4	56.3	83.6	58.9	52.8
YOLOv9-L [60]	-	200	43.1	192.5	40.1	61.0	85.2	60.1	56.5
YOLOv9-X [60]	-	200	54.7	284.9	35.8	61.6	82.9	63.6	57.2
YOLOv10-S [61]	-	200	8.3	24.4	37.3	55.7	78.2	55.4	50.5
YOLOv10-M [61]	-	200	16.4	63.6	37.6	56.8	79.5	57.1	51.9
YOLOv10-L [61]	-	200	25.5	118.4	38.9	57.4	83.2	58.3	55.9
YOLOv10-X [61]	-	200	32.5	166.5	40.2	55.2	84.7	58.6	52.8
YOLOv11-L [55]	-	200	25.3	85.9	35.7	56.7	83.6	60.3	57.1
YOLOv11-X [55]	-	200	54.8	191.3	39.5	59.1	84.0	59.9	53.4
YOLOv11-S [55]	-	200	12.6	24.7	40.6	58.4	82.5	61.3	55.3
YOLOv11-M [55]	-	200	21.2	67.6	32.3	56.7	83.1	62.7	53.3
YOLOv12-S [62]	-	200	10.4	24.7	35.8	55.7	81.4	58.3	53.6
YOLOv12-M [62]	-	200	21.1	67.4	38.2	56.8	82.5	61.9	54.2
YOLOv12-L [62]	-	200	26.4	92.2	40.8	59.1	82.9	63.4	57.8
YOLOv12-X [62]	-	200	58.2	197.4	42.1	58.6	84.7	63.1	58.5
Transformer Based									
Anchor-DETR [40]	R50	50	41.1	202.9	32.9	52.5	80.1	55.3	50.1
Anchor-DETR [40]	R101	50	79.5	231.7	33.4	51.2	79.1	56.0	51.2
Conditional-DETR [63]	R50	50	34.4	74.4	32.6	57.5	81.0	56.9	48.3
Conditional-DETR [63]	R101	50	51.3	290.0	33.9	55.2	81.6	56.6	48.9
Deformable-DETR [37]	R50	50	55.0	192.6	37.3	56.8	80.1	58.5	51.6
Deformable-DETR [37]	R101	50	65.9	270.9	35.5	56.7	80.2	60.5	53.9
DAB-DETR [39]	R50	50	63.9	192.2	34.1	54.9	79.0	59.6	53.1
DAB-DETR [39]	R101	50	73.8	254.9	31.1	53.8	78.3	55.3	49.4
DN-DETR [41]	R50	50	56.8	201.9	34.6	54.3	84.9	60.5	51.3
DN-DETR [41]	R101	50	77.3	284.5	39.5	56.8	82.4	58.3	57.6
Focus-DETR [42]	R50	36	60.2	196.5	32.6	60.3	84.8	62.2	56.5
Focus-DETR [42]	R101	36	74.2	274.6	33.1	56.5	77.4	54.2	47.9
DINO-4scale [34]	R50	12	60.1	191.4	34.7	54.7	79.3	58.2	50.5
DINO-4scale [34]	R101	12	76.3	214.8	38.6	56.5	82.6	58.5	56.8
SC-RTDETR [44]	R50	50	60.6	156.3	35.8	55.5	81.4	55.3	51.8
SC-RTDETR [44]	R101	50	81.1	183.5	31.4	57.9	83.2	60.4	52.2
ARS-DETR [45]	R50	50	50.5	146.4	38.7	58.6	80.1	56.4	53.3
ARS-DETR [45]	R101	50	82.7	171.7	37.8	58.2	81.4	59.9	56.8
PSWP-DETR [43]	R50	50	94.8	171.9	33.2	54.2	78.4	59.8	51.1
PSWP-DETR [43]	R101	50	149.5	253.3	35.7	57.7	78.9	58.6	52.8
OWRT-DETR [56]	R50	100	35.3	155.1	38.3	52.9	80.3	58.6	51.6
OWRT-DETR [56]	R101	100	59.8	275.3	39.4	58.8	78.6	59.3	55.1
GMODet [64]	R50	50	42.5	174.9	41.7	59.5	83.6	64.1	56.2
GMODet [64]	R101	50	101.1	287.4	37.7	55.2	78.9	58.8	56.8
SED-DETR [38]	R50	50	50.4	64.8	31.2	56.1	78.2	58.2	48.8
SED-DETR [38]	R101	50	60.9	84.2	33.1	57.8	77.3	54.4	50.6
RT-DETR [46]	R50	72	49.1	140.3	38.5	49.2	76.5	57.3	50.3
RT-DETR [46]	R101	72	78.2	263.6	38.8	59.7	78.7	58.6	55.2
RSRT-DETR (Ours)	R50	100	35.5	107.2	39.1	57.2	83.9	61.2	57.6
RSRT-DETR (Ours)	R101	100	67.4	205.8	41.0	54.9	82.3	58.1	55.6

from 34.7% to 35.8%. These experimental results effectively verify that PDSF Attention improves the model's sensitivity to small targets in complex contexts, thereby enhancing overall detection performance.

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TABLE VI
ABLATION STUDY OF RSRT-DETR COMPONENTS ON DOTA-V1.0 DATASET

Group	Components					Performance Metrics				
	PDSF	MS-HFNet	HSFN			AP_S	AP_M	AP_{50}	AP_{75}	$AP_{50:95}$
			MGSAL	ARGA	PESM					
1	✗	✗	✗	✗	✗	34.7	30.2	64.4	39.2	39.8
2	✓	✗	✗	✗	✗	35.8	30.4	66.9	39.6	40.3
3	✗	✓	✗	✗	✗	35.7	30.5	67.3	39.1	40.9
4	✗	✗	✓	✗	✗	35.5	30.1	66.9	38.3	39.6
5	✗	✗	✗	✓	✗	34.9	30.6	65.6	39.6	41.6
6	✗	✗	✗	✗	✓	35.1	30.3	67.4	39.2	41.2
7	✗	✗	✓	✓	✓	36.0	31.2	68.2	40.6	41.3
8	✓	✓	✗	✗	✗	35.9	30.7	67.8	39.5	41.2
9	✓	✗	✓	✓	✓	35.6	30.9	68.3	39.5	41.0
10	✗	✓	✓	✓	✓	36.2	30.5	68.6	40.1	41.9
11	✓	✓	✓	✓	✓	**36.3**	**30.6**	**69.3**	**40.9**	**41.1**

2) MS-HFNet: As shown in Table VI (Group 3), the introduction of MS-HFNet alone enhanced all evaluation indicators. Most notably, it improved the AP_{50} from 64.4% to 67.3%, representing a significant gain of 2.9%. This demonstrates that modeling high-order spatial correlations through hypergraphs effectively enhances the network's perception ability, even without other modules. Furthermore, when combined with PDSF Attention, the potential of MS-HFNet is fully exploited, leading to further improvements in all indicators, particularly AP_{50} and $AP_{50:95}$.

3) HSFN: As shown in Table VI, we analyzed the impact of HSFN by incorporating it into the baseline. The results show that HSFN provides the most substantial individual contribution. Specifically, when HSFN was introduced alone (Group 7), the model's AP_{50} improved by 3.8%, and small-object detection accuracy (AP_S) increased by 1.3%. Thanks to its refined interaction of cross-scale features extracted from the backbone network, it significantly boosts all performance metrics. Furthermore, HSFN exhibits strong compatibility, maximizing the model's performance when integrated with other proposed modules.

4) Summary of Ablation: Through the ablation experiment, we quantified the contribution of each module: HSFN contributes the most to the final accuracy (+3.8%), followed by MS-HFNet (+2.9%) and PDSF (+2.5%). These three modules are inseparable from the powerful extraction ability of RSRT-DETR for dense small targets, and they synergistically improve the robustness and accuracy of the model.

F. Grad-CAM Map Analysis

In order to more intuitively demonstrate the attention distribution characteristics of RSRT-DETR in different remote sensing scenarios, we introduced the gradient weighted class activation map (Grad-CAM) [65]. Its core idea is to estimate the spatial significance of different locations in the image through gradients. We performed a thermogram visualization of RSRT-DETR versus other object detection models on three datasets. Fig. 15 shows the attention focus of RSRT-DETR and other object detection models in different scenarios of three datasets. It can be found that in the heat map comparison of the three datasets, the parts focused by RSRT-DETR are more accurate. Thanks to the precise capture of edge information by PDSF Attention, and the combination of MS-HFNet improves the model's ability to extract small-target features, while HSFN performs secondary refining of feature information, allowing RSRT-DETR to achieve accurate focus on dense small targets. Not only does it effectively improve the detection accuracy, but also due to the clever design of these modules, RSRT-DETR has achieved lower computing overhead.

TABLE VII
FPS COMPARISON ON JETSON NANO (DOTA V1.0, SAME TRAINED WEIGHTS AS IN THE MAIN COMPARISON)

Model	FPS	Model	FPS	Model	FPS
Faster-RCNN	18	YOLOv12-X	157	DN-DETR	48
YOLOv5-X	112	EfficientDet	51	GMODet	113
YOLOv8-X	124	Anchor-DETR	32	SED-DETR	131
YOLOv9-X	119	Conditional-DETR	26	RT-DETR	143
YOLOv10-X	153	Deformable-DETR	37	RSRT-DETR	138
YOLOv11-X	163	DAB-DETR	40		
OWRT-DETR	138	Focus-DETR	79		
DINO-4scale	59	RTDETR-RCGA	126		
ARS-DETR	114	PSWP-DETR	104		

G. Actual Performance Testing of Edge Devices

While RSRT-DETR demonstrated promising performance in initial experiments, real-time inference speed FPS is a critical metric for drone-based detection tasks. The achievable FPS depends largely on the hardware of the deployment platform, as most UAVs possess limited onboard computational resources. To ensure the model fits these constraints, we

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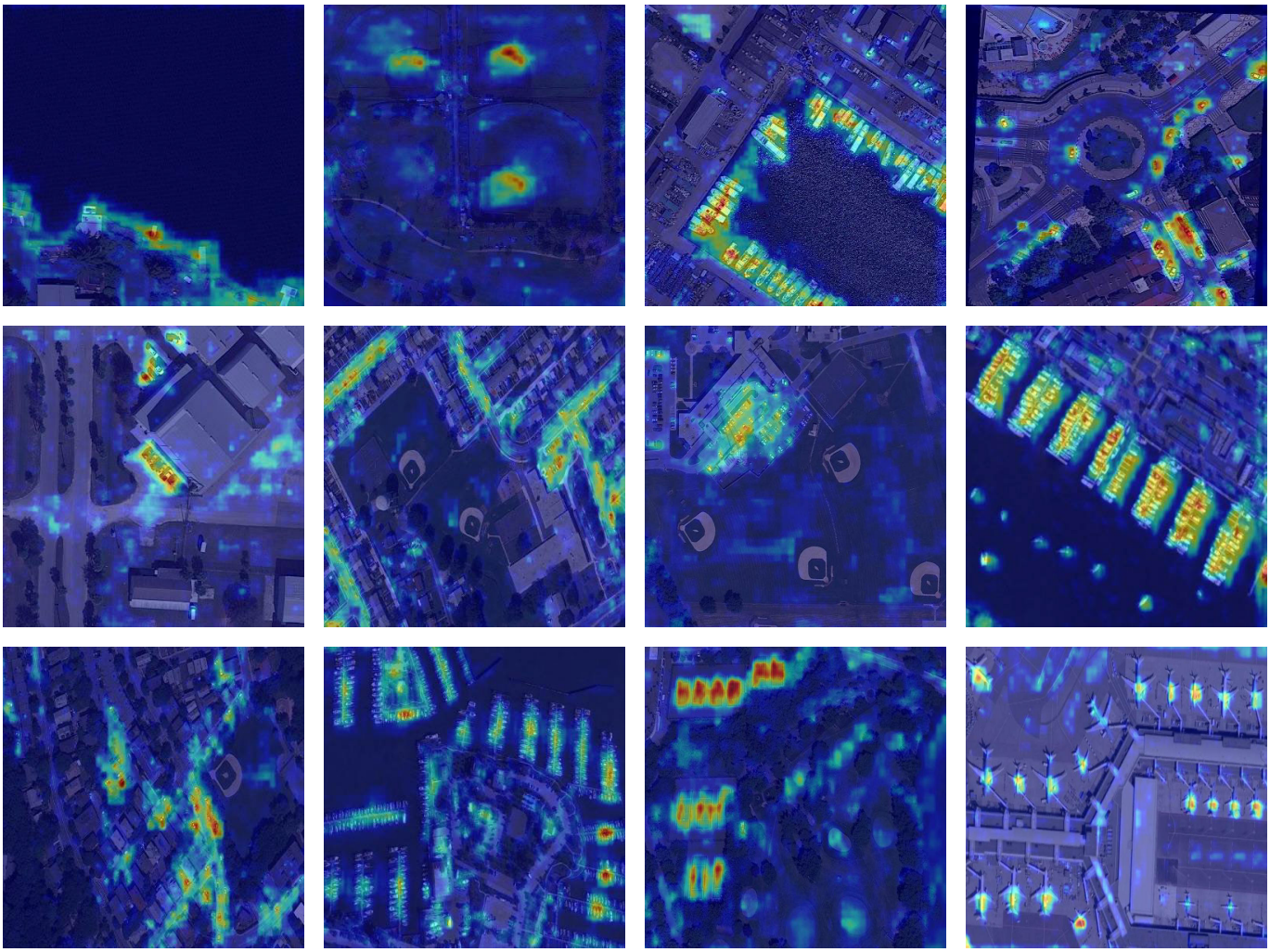


Fig. 15. Grad-CAM is used to visualize the RSRT-DETR processed images, where the brighter the color indicates the higher the attention the model has to the area.

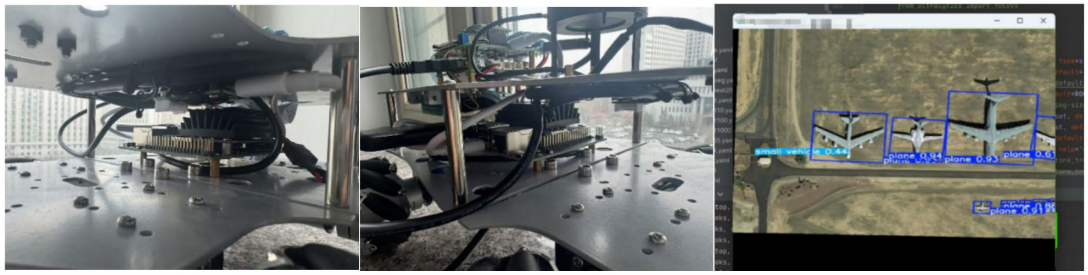


Fig. 16. Deployment and detection result diagram of edge devices. (a) Left-side view of the deployment platform. (b) Right-side view of the deployment platform. (c) Final test result graph.

incorporated specific lightweight strategies, including channel compression in the MS-HFNet and parameter sharing within the HSFN, to minimize computational redundancy. Furthermore, the architecture utilizes standard operators, facilitating efficient deployment via ONNX and TensorRT (FP16) optimization pipelines. To evaluate the deployment feasibility of RSRT-DETR on such edge devices, we tested it alongside several comparative models on an NVIDIA Jetson Nano board, a platform representative of typical UAV hardware. This setup simulates a realistic UAV hardware environment for inference testing [see Fig. 16(a) and (b)]. The test dataset was DOTA v1.0, and all inference weights were obtained from the models trained in the comparative experiments. The experiments show that RSRT-DETR maintains a high detection speed on resource-constrained edge devices, demonstrating good deployment practicality without significant performance

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TABLE VIII
FLOPs ESTIMATION AND DISASSEMBLY OF RSRT-DETR (R50) COMPONENTS

Component	FLOPs(G)	Percentage(%)	Description
Backbone(ResNet50)	78.5	74.7%	Pre-trained feature extractor
PDSF Attention	3.5	3.3%	Lightweight attention with Mona adapter
MS-HFNet	6.5	6.2%	Hypergraph modeling with channel compression (C/8)
HSFN	5.0	4.8%	Multi-scale fusion with parameter sharing
Transformer Decoder	6.5	6.2%	Sparse query updating
Prediction Head	5.1	4.8%	Detection heads and post-processing
Total	105.1	100%	Total Inference FLOPs

compromise. These results validate the model's practical value for resource-limited platforms and its suitability for remote sensing object detection tasks. Detection results from the Jetson Nano platform are presented in Fig. 16(c) and Table VII. The calculation amount estimation and disassembly of the RSRT-DETR (R50) component are presented in Table VIII.

V. CONCLUSION

This article proposes a new remote sensing object detection model, RSRT-DETR, based on DETR to address key challenges, such as low resolution, low proportion of small targets, multiscale dynamic changes, and complex background interference in the remote sensing images of drones. Specifically, it contains several innovations. First, PDSF Attention uses dual-path attention decomposition and dynamic normalization strategies to reduce computational redundancy while enhancing the detailed response to small targets. Second, the MS-HFNet network based on hypergraph modeling effectively captures the spatial correlation between multiscale targets. Finally, HSFN realizes the optimized fusion of cross-scale features through multilevel interactive modules. We conducted experiments on five datasets: DOTAv1.0, VisDrone2019, NWPU VHR-10, DIOR, and HIT-UAV. The experimental results show that the detection performance of this method is better than the current mainstream methods and SOTA models, effectively proving its detection ability of dense small targets in complex contexts. Despite showing superior performance in multiple benchmarks, there are still some aspects that can be optimized, which can be summarized into the following three points.

1) RSRT-DETR can further optimize memory utilization and detection speed.
2) Under extreme lighting conditions, the generalization ability of RSRT-DETR needs to be verified.
3) The existing research is mainly based on RGB mode data and has not fully integrated multispectral/hyperspectral information.

In future work, we will focus on exploring the fusion strategy of lightweight architecture design and multimodal features: on the one hand, we can achieve an independent balance between computing resources and detection accuracy based on the characteristics of edge computing platforms, and on the other hand, we will establish a cross-modal feature enhancement mechanism based on physical imaging models. These improvements will promote the development of drone detection systems in a more efficient and robust direction and provide more reliable technical support for practical applications, such as smart cities and environmental monitoring.

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